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Rational design of disordered proteins for sequence–function investigation

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Despite lacking a stable three-dimensional structure, intrinsically disordered protein regions (IDRs) are ubiquitous across all kingdoms of life and have essential cellular roles[1]. While rational design of folded proteins has seen substantial recent progress[2], our ability to design IDRs remains more limited[3]. Here we present GOOSE (Generate disOrdered prOteins Specifying propErties), a comprehensive computational framework for the rational design of IDRs. GOOSE's versatility and throughput enable us to design and test thousands of IDR sequences to reveal distinct sequence-to-function relationships. Using GOOSE to explore these relationships, we examine how sequence properties influence IDR structural ensembles in cells, design IDRs that respond to structural changes associated with cell volume decrease, create scaffold IDRs that self-assemble and recruit specific clients, and design novel IDRs that protect cells from desiccation. Our work uses rational sequence design as a powerful method for exploring function in IDRs and provides a versatile tool for designing functional disordered proteins.

IDRs are found in more than 70% of human proteins and have a variety of essential roles[1]. IDRs exist as a dynamic collection of rapidly interconverting and structurally distinct conformations[4] (Fig. 1a). Although structurally disordered, IDRs still possess sequence-encoded intramolecular and intermolecular interactions that define biases in structural ensembles, and mounting evidence suggests that these interactions are important drivers of molecular function across various cellular processes[1,5].

Despite their importance, systematically testing the relationship between IDR sequence and molecular function remains challenging. For folded domains, the encoding of specific three-dimensional structure in amino acid sequence has enabled the mapping of function from structure at a proteomic scale and the engineering of de novo functional proteins[2]. By contrast, our ability to understand function from IDR sequence and to use those principles to enable the design of functional disordered proteins is still emerging[1,3].

Pioneering work mapping IDR sequence-to-function relationships across various systems has enabled an alternative path forward: one in which sequence properties can be modulated to investigate sequence-to-function mappings[5–11]. Although highly effective in defining the contours of sequence–function relationships for IDRs, doing so at the scale of hundreds or thousands of sequences presents a considerable design challenge. These limitations reflect that fundamentally distinct paradigms for mutation and design are required to discern the molecular underpinnings and functional importance of IDRs (Fig. 1a).

Here we present GOOSE—a computational framework for the rational design of IDRs (Fig. 1b). GOOSE enables rapid design of de novo synthetic IDRs and variants of provided sequences, facilitating broad exploration of sequence space (Fig. 1b,c). GOOSE can design IDRs by sequence properties (such as amino acid composition, charge, hydrophobicity and charge patterning), conformational properties, chemically specific intermolecular interactions or any arbitrary design constraint (such as specific three-dimensional conformational ensembles[12]) (Fig. 1d). For built-in design objectives, GOOSE can generate thousands of sequences per minute (Fig. 1c and Extended Data Fig. 1b–d). Furthermore, GOOSE has no specific hardware requirements, making it broadly accessible. In brief, GOOSE enables fast and easy library-scale design of IDRs to systematically investigate sequence-to-function relationships.

Here we illustrate different design modalities and showcase the power of sequence design to achieve bespoke IDR functionalities. First, we use Förster resonance energy transfer (FRET) microscopy to examine the relationship between sequence properties and IDR ensemble in living cells. We then show how IDR ensemble relates to subcellular localization and sensitivity to cellular perturbations by designing IDRs with specific ensemble dimensions. Next, we demonstrate that GOOSE can design IDRs with specific intermolecular interactions by generating sequences that self-assemble or co-assemble with other designed sequences. Finally, we use GOOSE to design a library of variants to investigate the relationship between sequence and desiccation tolerance in yeast. Taken together, our work highlights how GOOSE can be used to gain insights into IDR sequence–function relationships.

IDR properties dictate ensemble in cells

Chemical interactions govern IDR conformational ensembles, dictating crucial structural features including global or local dimensions[13], accessibility of binding motifs[7], excluded volume effects and

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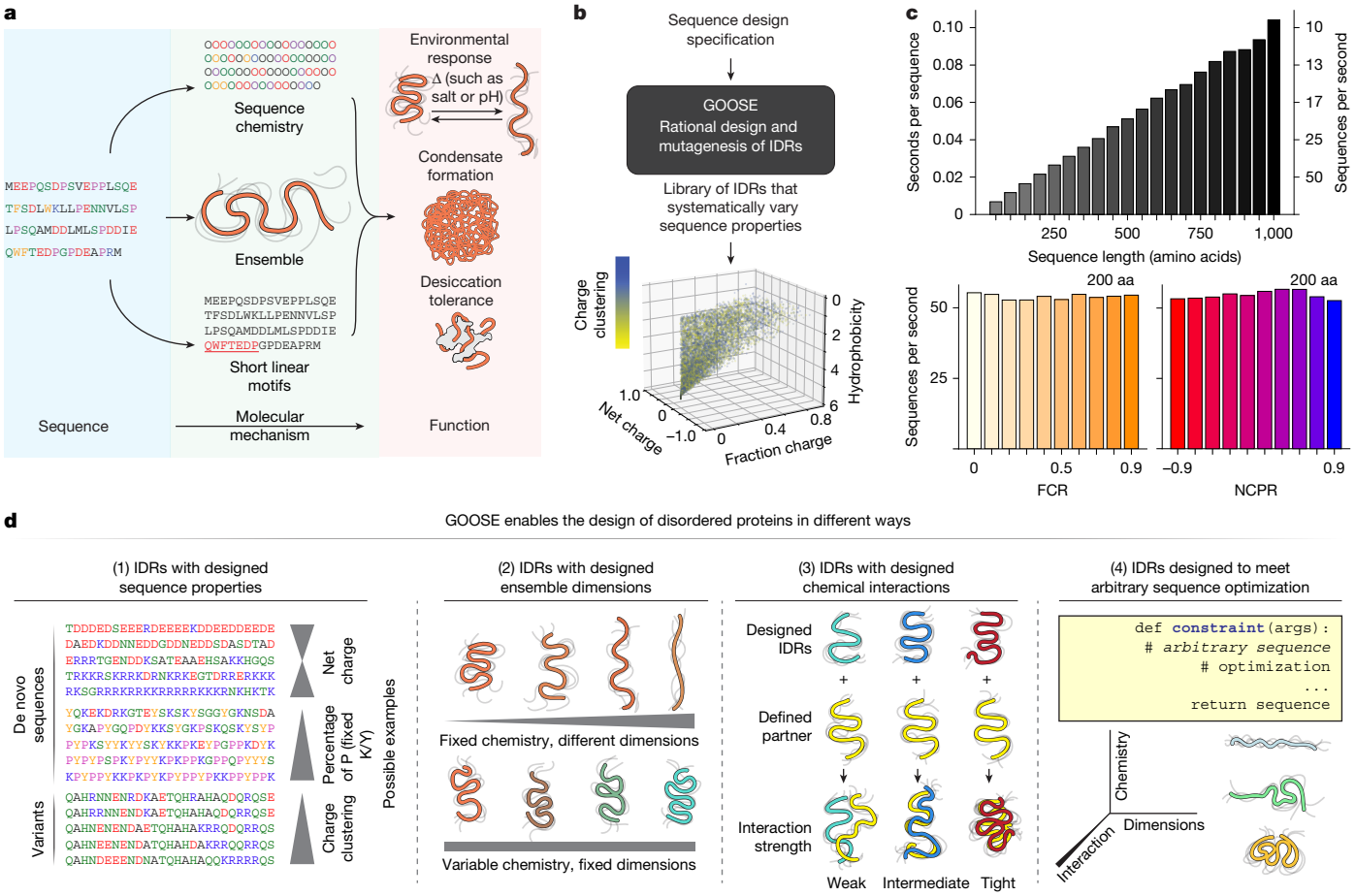

Fig. 1 | Overview of IDR function and a subset of GOOSE design functionality. **a**, IDR function is driven by a combination of short linear motifs, ensemble properties and sequence chemistry. **b**, GOOSE can design disordered proteins that systematically vary properties such as hydrophobicity, κ (charge clustering), FCR and net charge per residue (NCPR). **c**, GOOSE enables rapid generation of IDR libraries, enabling high-throughput exploration of sequence–function relationships. **d**, GOOSE can generate IDRs with specific sequence properties, ensemble properties, chemically specific intermolecular interactions or arbitrary design objectives. aa, amino acids; K, lysine; P, proline; Y, tyrosine.

conformational entropy[14,15], and multivalent binding cooperativity[16]. Ensemble-averaged dimensions, quantified by metrics such as the radius of gyration and the end-to-end distance, result from the balance between attractive and repulsive interactions, which can be intramolecular (dictated by sequence) and intermolecular (dictated by the chemistry of surrounding molecules).

While in vitro and in silico work has established core heuristics for sequence-to-ensemble relationships, systematic studies in cells are scarce[4,5,17]. To examine these principles in situ, we used GOOSE to generate 32 de novo IDRs designed to enable pairwise comparisons that differ by a single sequence property, each 60 residues in length (Fig. 2a and Extended Data Fig. 2). This length was chosen as a length scale that previous studies established as being capable of encoding diversity in the conformational ensemble[6], relevant to molecular function[7,18] and experimentally feasible in our system[19]. To measure the ensemble-averaged dimensions of these sequences in live cells, each IDR was flanked by a FRET donor–acceptor pair. Here, the FRET efficiency (E_f) functions as an inverse proxy for end-to-end distance: compact IDRs have a high E_f and expanded IDRs have a low E_f (refs. 4,19) (Supplementary Fig. 1; limitations are described in the Discussion).

We found that most rules elucidated in vitro or in silico hold true in cells. For sequences with a net charge (+0.3 or −0.3) and evenly distributed charges, increasing the fraction of charged residues (FCR) made ensembles more expanded (Fig. 2b–d), consistent with previous in vitro work[20–22]. Given that clustering of charged residues can impact ensemble behaviour[6], we next examined similar sequences with clusters of charged residues. For these sequences, we expected that increasing the FCR would drive compaction, which we observed for sequences with a net negative charge (Fig. 2e (left)). However, charge-clustered sequences with a net positive charge showed expansion after increased FCR (Fig. 2e (right)), a result that we speculate may be driven by interactions with other cellular components (Discussion).

To examine the environmental effects on ensemble structure, we performed coarse-grained simulations on these net-positively charged sequences (Extended Data Fig. 3a,b). These simulations reveal that polycationic sequences with evenly distributed charges show similar behaviour in silico and in cell (Extended Data Fig. 3a), whereas, after FCR increase, sequences with charge blocks show no significant change in silico but expansion in cell (Extended Data Fig. 3b), implying that additional factors in the cell influence ensemble structure.

We next examined how the FCR of net-neutral sequences influences IDR ensemble. For sequences with evenly distributed charged residues, ensemble dimension increased with increasing FCR (Fig. 2f (left)). However, while we expected that increasing FCR would drive compaction for net-neutral sequences with clustered charges, we observed both compaction and expansion (Fig. 2f (right)). We speculate that this difference may be tied to differences in proline content (Extended Data Fig. 3c–g) owing to proline-imparted backbone stiffness and/or favourable solvation energetics, consistent with previous work[20].

We next compared sets of three sequences where each set had equivalent FCRs (0.3 or 0.6), varying charge (−0.3, 0 or +0.3), and either evenly distributed or clustered charged residues (Fig. 2g). In silico and in vitro

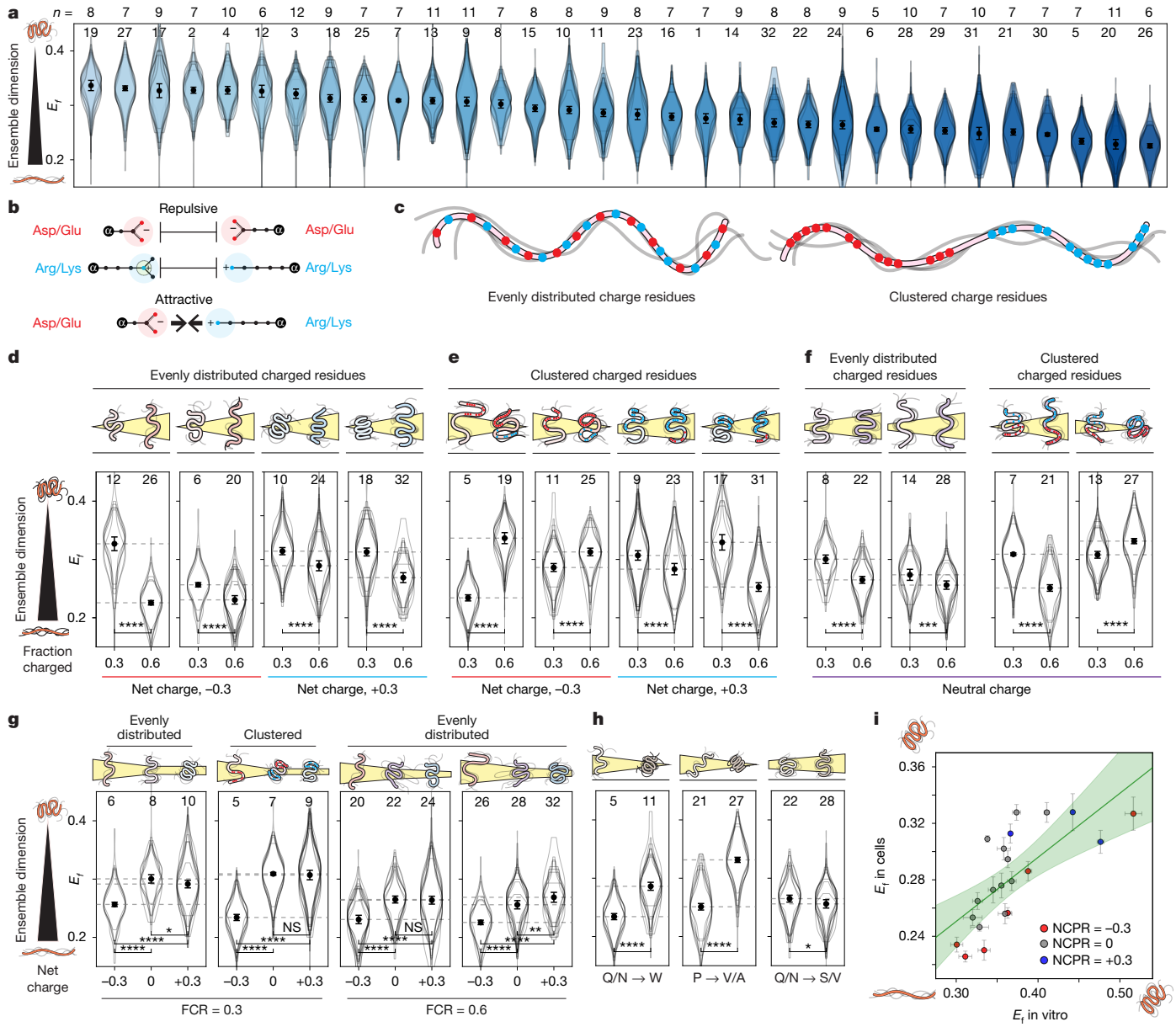

Fig. 2 | Sequence influences IDR dimensions in live cells. a, E_f values were obtained from live-cell microscopy for all 32 designed sequences (second row of values). In all of the violin plots in in Figs. 2 and 3, each of the overlaid violins represents a biological repeat with $n > 25$ cells, with the number of biological repeats shown at the top. For all in-cell metrics, including violin plots, data are mean ± s.d. of biological repeat median values. The significance between constructs was determined using unpaired two-sided Student's *t*-tests calculated on the well median values for each construct, then corrected for multiple testing using Benjamini–Hochberg correction; $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$. NS, not significant. **b**, Charged residues of the same polarity are repulsive, while charged residues of opposite polarity are attractive. **c**, Schematic of evenly distributed (left) or clustered (right) charged residues. **d**, The E_f for IDRs with evenly distributed charges and a net charge of −0.3 or +0.3. **e**, The E_f for IDRs with clustered charges and a net charge of −0.3 or +0.3. **f**, The E_f for IDRs with net neutral charge, increasing the FCR from 0.3 to 0.6. **g**, The E_f for IDRs with different charge patterning and FCR, titrating the net charge from −0.3 to 0 to +0.3. **h**, The E_f for sequences with non-charged sequence changes. **i**, Comparison of in-cell versus in vitro E_f values. Residues with a net charge of −0.3 are shown in red, and those with a net charge of +0.3 are shown in blue. The green line is a linear fit to the data, with the 95% confidence intervals shown by the shaded area. For in vitro data, data are mean ± s.d. across 3 biological replicates.

studies predict that the ensemble dimensions of IDRs with the same charge but opposite signs should be similar[20,22,23]. However, our results found that net-positive and net-neutral sequences showed similar compaction, whereas net-negative sequences were more expanded. This holds true at an FCR of both 0.3 and 0.6 and clustered or evenly distributed charges. Notably, introducing aromatic residues into sequences with evenly distributed charges can override this expansion (Supplementary Fig. 2).

To further differentiate sequence-intrinsic effects from those dependent on the intracellular environment, we performed in vitro FRET (Fig. 2i). Note that, owing to a failure to express sequences with a large number of positively charged residues heterologously, we examined only 21 of 32 total sequences in vitro. The in vitro FRET yielded generally good correlation with in cell FRET (Pearson's $r = 0.70 \pm 0.04$; Fig. 2i). In vitro results confirmed expected behaviours: increasing the FCR in evenly distributed charges expanded ensembles (Extended Data Fig. 4a) whereas introducing aromatic residues drove compaction (Extended Data Fig. 4b). Importantly, sequences that contained some positive charges and displayed unexpected compaction in cells showed mild or no compaction in vitro (Extended Data Fig. 4c), consistent with

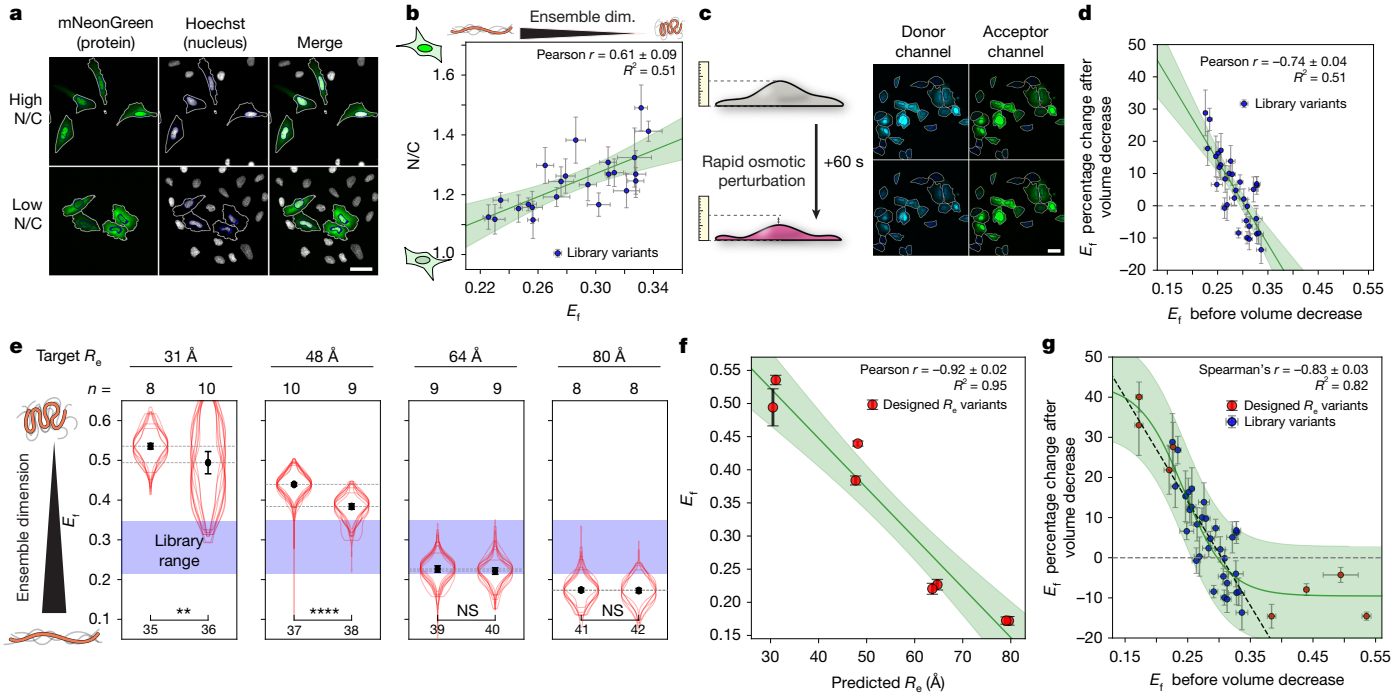

Fig. 3 | Ensemble dimensions affect both nuclear localization and conformational response to osmotic shock. a, Representative images displaying differences in nuclear localization as determined by the ratio of fluorescence in the nucleus and the cytoplasm (N/C). Cell contours are shown with a white outline. Scale bar, 50 µm. **b**, The correlation between the E_f and the N/C for each construct. The green line is the linear fit of the data. Shaded green regions are the 95% confidence intervals of the fit. dim., dimension. **c**, Schematic of osmotic perturbation and representative images of FRET donor and acceptor channels before and after osmotic perturbation. Scale bar, 50 µm. **d**, The change in E_f after hyperosmotic perturbation versus the E_f before perturbation. The green line is the linear fit to the data. **e**, The E_f distributions for designed R_e sequences, with design target above each construct pair. The blue-shaded region represents the range of median E_f values for the library shown in Fig. 2a. Each of the overlaid violin plots represents a biological repeat with $n > 25$ cells; the number of biological repeats is shown at the top. Data are mean ± s.d. of the median values. Significance between constructs was determined using unpaired two-sided Student's *t*-tests between biological repeat median values per construct; *P* values were corrected for multiple testing using Benjamini–Hochberg correction. **f**, The relationship between E_f and R_e predicted by ALBATROSS. The green line is the linear fit of the data. **g**, The change in E_f as in **d**, but with the designed R_e sequences included. The green line is the data fit to a variant logistic growth equation, and the dashed black line is the linear fit from **d**. See Fig. 2a for description of markers and error bars.

our hypothesis that the unexpected compaction is owing to interaction with other cellular components[24]. Furthermore, net-negative sequences were more compact in vitro than in cells (Fig. 2i).

Taken together, our results in cells are consistent with previous in vitro and in silico observations with a few exceptions—notably, charge clustering can have complex effects on chain dimensions, and net-positive sequences are unexpectedly compact.

IDR ensemble influences in cell behaviour

We next examined the relationship between the average ensemble of our IDRs and their nuclear versus cytoplasmic localization (Fig. 3a). Molecular size can influence nuclear import, with smaller proteins diffusing more freely into the nucleus[25]. We found that as ensembles compact their nuclear localization increases (Pearson's $r = 0.61 \pm 0.09$) (Fig. 3b). That said, we note that factors in addition to size also influence nuclear localization (Discussion).

IDRs exhibit a variety of ensemble-dependent responses to crowding as a function of crowder size in vitro and in cells[19,26]. Generally, increased crowding restricts space and drives ensemble compaction[27]. We therefore sought to examine whether there was a relationship between ensemble and sensitivity to crowding. To test this, we added mannitol to the imaging medium, causing a rapid reduction in cell volume and increased cellular crowding[28]. FRET imaging before and after mannitol addition resulted in compaction of expanded sequences and had little impact on compact sequences, consistent with our expectations (Pearson's $r = -0.74 \pm 0.04$) (Fig. 3d). However, a few sequences expanded under increased crowding, which may be owing to increased interactions with binding partners or increased concentration of solutes (for example, ionic species) that mask attractive intramolecular interactions[29].

To further examine the relationship between ensemble and sensitivity to crowding, we used GOOSE to design eight 60-residue sequences that systematically titrate predicted end-to-end distances. Notably, these sequences extend the range of dimensions beyond those of our initial library (Fig. 3e and Supplementary Table 1). Under basal conditions, the new sequences showed the expected correlation between E_f and the target dimensions (Fig. 3f). We next tested these sequences in our crowding assay and found that highly expanded sequences exhibited the greatest compaction after crowding. Moreover, for compact sequences, a plateau in E_f change was observed, indicating that highly compact sequences cannot be further compacted by crowding (Fig. 3g). These results showcase our ability to design sequences with increased sensitivity to crowding and to design IDRs with specific responsiveness to intracellular environmental conditions.

Designing IDRs that self-assemble

IDRs can interact with other biomolecules through chemically specific interactions whereby interaction is driven by complementary local or global sequence chemistry[1] (Fig. 1a). We recently developed FINCHES—a computational tool that can calculate a mean-field intermolecular interaction score (ε) between two IDRs[30] (Fig. 4a). Here, negative ε values indicate attraction and positive ε values indicate

repulsion between IDRs. To establish the validity of ε in the cellular context, we reanalysed our sequences reported in Fig. 3 and compared ε with the basal E_f. Previous work has established a relationship between IDR dimensions and self-interaction strength[31]. Consistent with this, we found a reasonable correlation between E_f and ε (Fig. 4b), suggesting that ε values can reflect IDR interactions in the cellular context.

Biomolecules can engage in homotypic interactions (for example, protein A with another protein A) or heterotypic interactions (protein A with protein B). We used intracellular self-assembly as a proxy for homotypic interactions, reasoning that IDRs with strong attractive homotypic interactions would be likely to self-assemble, whereas those with repulsive interactions would not. Although there are caveats associated with such an approach (Discussion), this logic has been used previously[32]. We designed a series of 200-residue IDRs with variable homotypic interaction strengths and transformed a fluorescent protein (FP)-tagged version of each into *Saccharomyces cerevisiae* (Fig. 4c,d and Supplementary Table 2). We then compared their subcellular localization to the FP alone (Fig. 4e). The FP-alone control and two self-repulsive IDRs all appeared diffuse (Fig. 4e (left)), whereas the two self-attractive IDRs displayed punctate localization (Fig. 4e (right)). These results showcase our ability to design IDRs that undergo self-assembly in cells; however, we cannot exclude the possibility that our designed IDRs are co-assembling with other cellular components[30].

To further test our ability to design IDRs with targeted chemical specificity, we designed 400-residue scaffold IDRs that self-assemble, along with various 200-residue clients that should be recruited to or excluded from the scaffold assembly (Supplementary Table 3). We first established that our scaffold IDRs self-assemble in the absence of clients (Extended Data Fig. 5a,b). Next, we co-expressed clients designed to be recruited to (Fig. 4f; client designs) or excluded from (Fig. 4g; not-client designs) scaffold assemblies (Extended Data Fig. 5c). These designs followed the expected behaviour, confirming our ability to design IDRs with desired chemically specific interactions.

Finally, we reasoned that a 400-residue client would show enhanced recruitment compared to a 200-residue client. Designing and testing a 400-residue client, we found approximately ten times greater client recruitment, despite the client being diffuse in the absence of the scaffold (Extended Data Fig. 5d). Indeed, across all designed clients for scaffold 1 and scaffold 2, we recovered expected partitioning behaviour, despite substantial variation in sequence between clients and across the two scaffolds (Fig. 4h,i and Supplementary Figs. 3 and 4). Overall, these results suggest that we can use GOOSE to design IDRs with the desired intermolecular interaction profiles.

IDR variant library for desiccation

Although multiple strategies confer desiccation tolerance in a range of organisms[33], the synthesis of protective IDRs is used across all kingdoms of life[34]. However, systematic exploration of the relationship between sequence properties and desiccation protection has been limited[35,36]. We used GOOSE to create a library of 2,300 variants based on 11 template sequences. These templates, both synthetic and naturally occurring, had a range of protective capacities against desiccation in *S. cerevisiae* (Extended Data Figs. 6 and 7a). For each template, variants were designed that systematically titrate sequence properties (for example, average hydrophobicity, net charge and global dimensions) (Fig. 5a, Supplementary Fig 5 and Supplementary Table 4). We transformed our library into *S. cerevisiae* and performed a sequencing-based desiccation-protection assay. For this, DNA sequencing was used to quantify the abundance of each variant in the yeast population before and after a desiccation, rehydration and regrowth cycle (Fig. 5b and Methods). We use the fold change (FC) in normalized read abundance before and after desiccation and regrowth to identify sequences that are protective, neutral or deleterious.

As an initial check, we subjected yeast transformed with either the library or an empty vector (EV) control to desiccation and then tracked subsequent regrowth, finding that yeast transformed with the library consistently grew faster than the yeast transformed with the EV control (Extended Data Fig. 7b). Then, by quantifying the FC across three experimental replicates (each with three biological replicates), we established the baseline impact of the templates on desiccation tolerance (Fig. 5c) and identified numerous variants that were deleterious or protective (Fig. 5d,e). As a negative control, the library included sequences with early stop codons that should display little to no effect in terms of protective capacity (Fig. 5f). Of 77 controls, only 6 (8%, compared with 27% for non-control sequences) showed protective effects (Fig. 5g), and even these had marginal FC values.

We next examined our data to identify sequence features associated with desiccation protection. Using a combination of sequence analysis and prediction packages[18,37], we identified 130 sequence features for each sequence (Supplementary Table 4). Focusing on the most protective ($n = 216$, adjusted P (P_{adj}) < 0.05, log[FC] > 0.5) or deleterious ($n = 63$, $P_{adj} < 0.05$, log[FC] < −0.5) sequences, we identified features that are significantly different between protective and deleterious sequences across all templates (Fig. 5h–k and Extended Data Fig. 8). Protective sequences showed higher alanine content (Fig. 5h), lower hydrophobicity (Fig. 5i) and increased helical propensity (Fig. 5j), all of which are consistent with reports on functional desiccation-protective proteins[38–41]. We also examined homotypic interaction strength using FINCHES, finding that protective sequences tended to be self-repulsive (positive ε) (Fig. 5k). This contrasts with recent observations suggesting that self-assembly of protective IDRs is associated with desiccation tolerance[42,43]. However, we note that protectants are hypothesized to self-assemble into vitrified phases in environments in which their concentrations are extremely high owing to a lack of water[36]. High self-attraction would prompt them to assemble even before the onset of desiccation, possibly hindering their function[44]. Finally, we compared our top protective sequences to the yeast proteome. This analysis revealed that these sequences are distinct outliers relative to endogenous yeast IDRs across many features that distinguish them from the bottom performers, including self-repulsion, helix probability and the fraction of alanine residues (Extended Data Fig. 9a).

We next quantified the correlation between protection and sequence features for each template (Supplementary Fig. 6a), excluding sequences within 7% similarity to the template. This revealed that, for most templates, decreasing the sequence complexity, increasing the disorder propensity and increasing the alanine fraction enhanced the protective capacity (Fig. 5l,m), consistent with the overall trends (Fig. 5h–k and Extended Data Fig. 8). While some parameters tuned the protection of all sequences in the same way (Fig. 5n), not all templates were equally tunable by the same parameter, and the protection of some templates was overall more tunable than others (Extended Data Fig. 9b). This supports the importance of the sequence background or context beyond any individual parameter in determining protective function[45]. Taken together, our results showcase the power of using GOOSE to design IDR libraries to identify the sequence features associated with IDR function. While our results were limited to a relatively small library, this approach can be extended and easily modified to create an iterative pipeline that produces potent functional IDRs tailored to specific functions in targeted environments.

Discussion and conclusion

IDRs constitute more than 30% of the human proteome and have critical roles in most aspects of eukaryotic biology[1]. Despite this, our understanding of the IDR sequence-to-function relationship is still emerging. This is in part owing to the difficulty of iteratively and systematically perturbing IDRs. For folded proteins, structural information provides a route to hypothesis-driven investigation. However, the lack

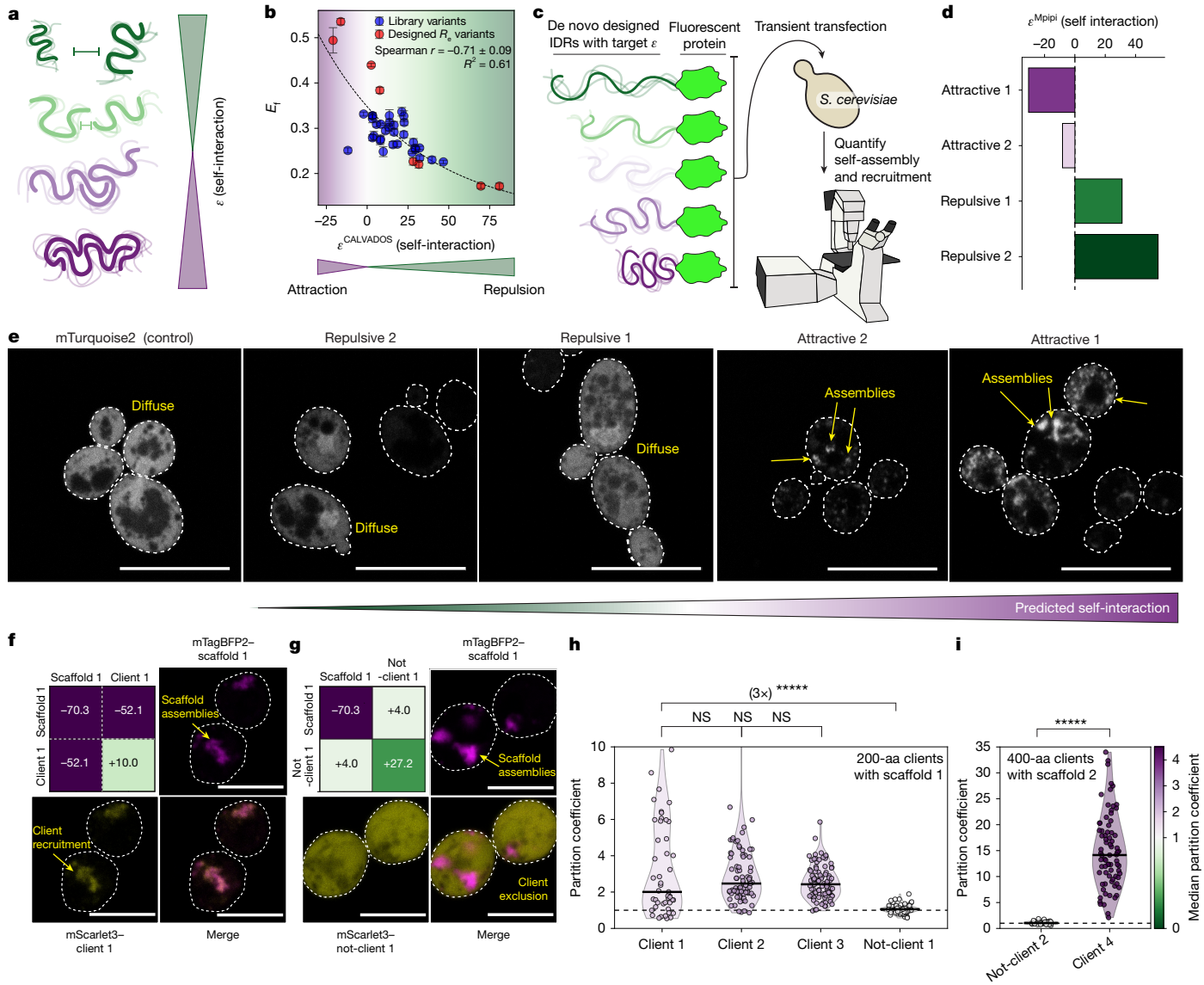

Fig. 4 | Design of IDRs with specific intermolecular interactions. a, FINCHES enables the calculation of a single, mean-field parameter that captures IDR–IDR-mediated interactions. **b**, Correlation of the self-interaction ε value and the measured E_i value (as reported in Figs. 2 and 3). **c**, Overview of the experimental setup for quantifying intracellular interaction. **d**, Self-interaction ε for 200-residue designed sequences. **e**, Microscopy analysis of self-assembly designs shown in **d** and a control (mTurqouise2). Scale bars, 10 µm. Representative images are shown of cells expressing the indicated sequence and were observed across at least two imaging sessions. **f**, Microscopy images for a system in which a self-assembling scaffold (scaffold 1) forms, and a second component was designed to be recruited to that scaffold (client 1). Scale bars, 5 µm. **g**, Microscopy images for a two-component system in which a self-assembling scaffold (scaffold 1) forms, and a second component was designed not to be recruited to that scaffold (not-client 1). Scale bars, 5 µm. For **f** and **g**, representative images are shown of cells expressing both constructs as observed across two imaging sessions. The results shown in **f** and **g** are quantified in **h**. **h**, Quantification of co-localization of 400-residue scaffold (scaffold 1) and 200-residue clients. The violin plots show the kernel density estimate across partition coefficients and are coloured by the median partition coefficient. The solid black lines show the median of each distribution, while black dashed line marks a partition coefficient of 1. Individual datapoints are shown for individual assemblies in different cells. Clients 1, 2 and 3 are statistically indistinguishable, whereas not-client 1 is significantly different from all other constructs. Statistical analysis was performed using pairwise two-sided Mann–Whitney *U*-tests with Holm–Bonferroni correction; exact *P* values are provided in Supplementary Fig. 3. $n = 47$ condensates from 13 cells across 7 *z* stacks (client 1), $n = 81$ condensates from 21 cells across 8 *z* stacks (client 2), $n = 91$ condensates from 31 cells across 13 *z* stacks (client 3) and $n = 71$ condensates from 23 cells across 15 *z* stacks (not-client 1). **i**, Analogous data for a 400-residue scaffold (scaffold 2) and a 400-residue client. Larger clients show substantially greater partitioning (around 10×), in agreement with expectations from enhancing multivalency. The 400-residue not-client 2 and 400-residue client 4 are significantly different from one another; statistical analysis was performed using pairwise two-sided Mann–Whitney *U*-tests with Holm–Bonferroni correction; exact *P* values are provided in Supplementary Fig. 4. $n = 37$ condensates from 21 cells across 7 *z* stacks (not-client 2) and $n = 77$ condensates from 19 cells across 6 *z* stacks (client 4). *****$P < 1 \times 10^{-6}$.

of a reference structure for IDRs raises a conceptual barrier relating to how we investigate sequence determinants of IDR function in the absence of a structural basis for function.

To address this, we introduce GOOSE, a tool for designing de novo and variant IDRs. GOOSE enables rapid and systematic generation of IDRs with varied built-in design objectives for key sequence features and properties that are important determinants of IDR function. Furthermore, GOOSE can use arbitrary design objectives to generate IDRs with bespoke sequence properties, thereby facilitating multi-objective optimization. Across several systems (in vitro, in yeast

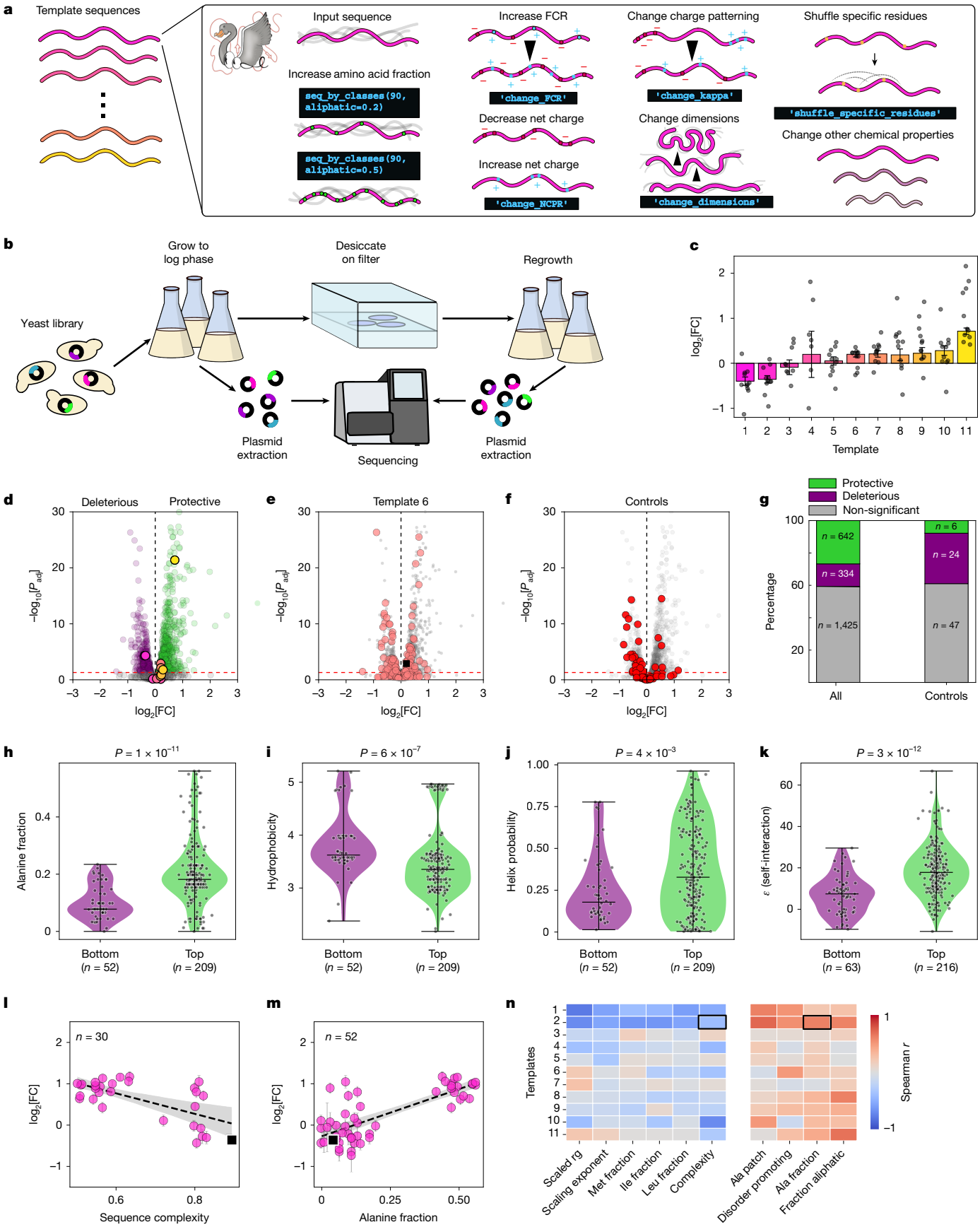

Fig. 5 | See next page for caption.

Fig. 5 | Rational design of desiccation protective IDRs. a, Experimental schematic showing how GOOSE was used to generate an IDR variant library. **b**, Schematic of the experimental pipeline. **c**, Template sequences organized by template number. The bar height represents the inverse-variance weighted mean $\log_2$[FC] from Supplementary Table 4. The error bars represent the inverse-variance weighted s.e.m. across three independent experiments. Individual points show paired $\log_2$[after/before] ratios per biological replicate, excluding zero read counts. **d**, The FC versus false-discovery-rate (FDR)-adjusted *P* value. $\log_2$[FC] values are the inverse-variance weighted means from three independent experiments. *P* values were calculated using two-tailed *t*-tests and corrected using Benjamini–Hochberg correction. The purple and green points indicate deleterious and protective sequences, respectively. **e**, The FC versus FDR-adjusted *P* value as in **d** for variants of a single template (black square). **f**, The FC versus FDR-adjusted *P* value as in **d** for library internal controls. **g**, Protection across 2,300 variants compared with internal controls. **h**–**k**, Comparison of the alanine fraction (**h**), hydrophobicity (**i**), helix probability (**j**) and self-interaction (**k**) of the most protective (top; green) and most deleterious (bottom; purple) sequences, as defined in Extended Data Fig. 8. The violin plots show the kernel density estimates; the horizontal lines indicate the median values and the whiskers extend to the minimum and maximum values. Exact *P* values, as calculated using two-tailed Mann–Whitney *U*-tests, are shown above each pair. The number of different sequences (*n*) is listed below the graphs. **l**,**m**, Desiccation protection versus sequence complexity (**l**) and the fraction of alanine residues (**m**) for template 2. Each point (*n*) is a variant; the error bars are as described in **c**. The wild-type sequence is shown as a black square. The dashed line shows the linear regression fit; the shaded band shows the 95% confidence interval. **n**, Spearman's *r* between desiccation protection and sequence features across templates. Red, positive correlation; blue, negative correlation. The black boxes highlight correlations shown in **l** and **m**. rg, radius of gyration (as computed using ALBATROSS).

and in human cells), we illustrate a range of use cases for GOOSE, revealing sequence-encoded determinants of IDR sensitivity, their capacity for homotypic and heterotypic intermolecular interactions and their ability to function as protective agents against environmental stress.

Examining sequence–ensemble relationships in cells, many of our results are consistent with established theory: increasing the net charge, the overall FCR and proline residues tends to expand the ensemble dimension (Fig. 2d–f and Extended Data Fig. 2c–g), whereas adding aromatic residues tends to lead to compaction (Fig. 2h). However, for positively charged tracks, the results are less clear. Increasing the net charge of sequences led to expansion in some cases (Fig. 2d), and had no effect (Fig. 2g) or led to compaction (Fig. 2g) in others. Furthermore, sequences with clustered positively charged residues became more expanded (Fig. 2e) or more compact (Fig. 2f) as FCR increased. This could in part be owing to positive charge clusters acting as nuclear localization signals, as many sequences containing them localize to nuclear puncta (Extended Data Fig. 10), consistent with previous work showing that clusters of positive charge is sufficient for recruitment into RNA-rich nuclear subcompartments[24,46]. Finally, our negatively charged sequences were generally more expanded in cells than in vitro (Fig. 2i). In summary, these results describe a cellular milieu in which positively charged biomolecules tend to interact with other components, and negatively charged biomolecules are expanded and more soluble, consistent with a growing body of work across a variety of organisms and systems[24,47].

In examining IDR behaviour in cells, we observed a high correlation between ensemble dimension and nuclear localization for our original 32-sequence library (Fig. 3b). However, after introduction of the eight additional end-to-end distance (R_e) -designed sequences, the correlation becomes weaker (Supplementary Fig. 7). This behaviour may reflect the complexity of subcellular localization: passage through the nuclear pore depends on a combination of factors, including the presence of a nuclear localization signal, chemical miscibility with the nuclear pore, ensemble dimensions and the availability of soluble protein for nuclear uptake[25,48]. When we sample the very distinct chemical space of sequences 35–42, we change much more than just the ensemble dimension. While one of the constructs designed to be compact (sequence 35) shows substantial nuclear enrichment, another (sequence 36) forms cytoplasmic assemblies, possibly limiting the pool of soluble protein available for nuclear import (Extended Data Fig. 10). These results highlight the complex interplay among competing factors, which can obfuscate mechanistic insights when physical models for the underlying processes are incomplete.

When testing the sensitivity of IDR ensembles to in-cell crowding (Fig. 3c), we identified a strong correlation between basal ensemble dimensions and sensitivity (Fig. 3d). By generating sequences to explicitly titrate predicted end-to-end distance (Fig. 3e), we confirmed that this correlation plateaus at ensemble extremes (Fig. 3g). This showcases the power of designing IDRs with specific global dimensions. Other phenotypes, such as diffusion rates, binding affinity and lifetimes, may also be modulated by ensemble dimensions and are therefore viable targets for the systematic design capacity offered by GOOSE. Furthermore, these results highlight the continuation of trends when ensemble dimensions are taken to extremes, into sequence space, where the chemistries and resultant ensembles are outside the biological norm.

Note that, for all FRET experiments, the terminal FPs could impact the ensemble—whether through IDR–FP or FP–FP interactions[4]. Nonetheless, we believe that these interactions do not dominate our conclusions: all designs were based solely on the IDR sequence. Overall agreement with predicted IDR dimensions (Fig. 3e), a high correlation with self-ε (Fig. 4b) and expected E_f shifts seen in pairwise sequence comparisons (Fig. 2) suggest that FP interactions are probably not a primary determinant of the observed signal.

After examining IDR ensembles in various contexts, we designed IDRs that undergo self-assembly in vivo and client IDRs targeted to those assemblies (Fig. 4). These assemblies are examples of biomolecular condensates—non-stoichiometric cellular bodies that concentrate specific biomolecules and exclude others; however, we emphasize that we make no claims regarding their composition or material state[49]. Across these experiments, we consistently observed the expected behaviour regarding self-assembly or recruitment to a target assembly. However, we note that, although assembly is a convenient readout of homotypic or heterotypic protein–protein interactions, it is not without limitations. These experiments were conducted in cells. Thus, our designed IDRs almost certainly interacted with additional off-target biomolecules that could influence self-assembly or client recruitment. Furthermore, we note that, although our ability to design scaffolds and clients targeted to them supports the possibility of using GOOSE to target clients to native condensates or to design condensates that recruit specific native proteins, we have not yet tested this. In principle, there is no reason that this could not be done, and it would be an interesting area for future exploration.

Finally, we used GOOSE to generate variants to examine the relationship between sequence properties and desiccation protection. Across the different templates used to generate our variants, there was substantial variation in designability (Extended Data Fig. 9b). Templates of synthetic origin or unrelated to desiccation protection offered little protection, and variants generated from these templates often did not improve protection. By contrast, templates associated with desiccation tolerance in other organisms (Extended Data Fig. 6) exhibited a greater dynamic range of desiccation protection across their variants. However, the specific sequence changes that improved protection were generally not consistent across the 11 templates, with the exception of a few features that were previously associated with desiccation-protective IDRs[50].

Notably, the overall effect measured as the FC in read counts before and after desiccation and regrowth may appear modest—a factor of eight at most. This largely reflects our experimental design, in which regrowth is limited to around 15–20 h after desiccation (Extended Data Fig. 7b). However, prolonging regrowth would drive FC values upward.

Nonetheless, we identified sequences that significantly affected protection in nine biological replicates collected across three independent experiments.

Predicting which IDRs are more or less designable a priori appears difficult, at least for desiccation tolerance. Using GOOSE to generate a library of variants identified which IDR templates were most designable and the sequence features associated with improved function. Although not done here, one could iteratively design and test libraries using GOOSE to continually optimize a sequence towards some objective function. This would reveal sequence-to-function relationships and enable the design of IDRs with strong functional effects.

Here we tested only a very small number of IDR sequence-to-ensemble or sequence-to-function relationships. However, there are many more areas that we believe GOOSE can be used to test hypotheses with respect to how IDR sequence drives function. In particular, recent studies examining IDRomes from a single organism have identified numerous relationships between IDR sequence and associated molecular function[18,30,51,52]. It would be interesting to use GOOSE to explicitly test some of these relationships in future studies.

Work from several groups has provided tools focused on the design of IDRs[3]. These methods include using iterative simulations[8] or machine-learning-based approaches focused on ensembles, interactions or generative design through language models[53–58]. We see these methods as complementary (or even synergistic) to GOOSE; while these approaches enable the generation of synthetic natural-looking IDRs, they are not designed to generate sequences that systematically titrate human-interpretable features for hypothesis testing. Furthermore, given GOOSE's SequenceOptimizer, emerging deep learning methods can be combined with GOOSE to turn predictive methods into design objectives[12].

Designing stable, soluble folded domains requires cooperative interactions among many residues and regions. For a folded domain to function—regardless of what it does—correct folding and solubility are unavoidable prerequisites. Given that functionally equivalent folded domains from across the kingdom of life often (though not always) are sequence similar, we (and others) subscribe to a model in which 'structure' encodes a tight coupling between sequence and function, with even small perturbations in the sequence that disrupt the structure having devastating consequences for function.

For IDRs, a different paradigm emerges. No unifying feature underlies IDR function. When assessed using alignment-based metrics, IDRs often appear poorly conserved, reflecting the empirical observation that many sequences can appear equally functional across different systems[45,59]. Taken at face value, this suggests that, for a given function, the design space for IDRs is large. As such, if sequence-to-function mapping for folded domains involves a hypothetical fitness landscape with one (or a small number) of deep wells, analogous landscapes for IDRs involve broad, roaming hills and valleys. In this analogy, the small, incremental sequence perturbations that are so informative for exploring folded-domain function may be poorly suited to exploring this landscape. By contrast, we propose that, for IDRs, investigating sequence-to-function relationships by systematically perturbing many sequences is more useful, at least initially. Rather than testing a few top candidates, we propose that IDR sequence-to-function relationships require large, high-throughput experiments that map these landscapes across tens, hundreds or thousands of sequences.

One route to do this may be to use evolutionary-scale large language models for generative design of synthetic IDR variants[53,60,61]. A non-overlapping and complementary strategy is presented in GOOSE, in which users define their own hypotheses about which sequence features matter and then design large libraries capable of simultaneously testing many competing hypotheses in parallel. By enabling the design of large (10,000+) sequence libraries in minutes to hours, GOOSE provides both the speed and the capabilities to match high-throughput genotype–phenotype mapping. Looking forward, this opens the door to broad-reaching investigations of IDR function in natural proteins, as well as a wide range of biotechnological applications.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-026-10849-1.

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Methods

About GOOSE

GOOSE is a software package for the design of disordered protein sequences. The sequences used in this Article were designed using the Python (v.3.9+) package GOOSE (https://github.com/idptools/goose, https://doi.org/10.5281/zenodo.20127102). GOOSE uses sparrow (https://github.com/idptools/sparrow/) to calculate sequence properties. Ensemble predictions used to design IDRs with a desired radius of gyration or end-to-end distance are obtained using ALBATROSS, as implemented in sparrow[37]. ALBATROSS is a deep learning tool for predicting ensemble-average IDR dimensions directly from sequence and was parameterized based on coarse-grained simulations performed with a modified variant of the Mpipi model[62]. GOOSE uses FINCHES (https://github.com/idptools/finches) to calculate predicted intra- or intermolecular interactions, where FINCHES makes use of both the Mpipi and CALVADOS forcefields[30,62,63].

GOOSE has two general design approaches: predefined generation and variation functionality and sequence optimization. Both approaches enable the creation of fully synthetic sequences or starting from a sequence of interest (Supplementary Fig. 5). For predefined functions, fully synthetic sequence generation enables one to specify various properties (hydrophobicity, FCR, net charge per residue (NCPR) or κ), fractions of amino acids, average ensemble properties and patterning properties (for example, how symmetrically positioned aromatic amino acids are). For predefined functions for variant generation (when you start with a sequence of interest), GOOSE has 18 different types of variant that can broadly be categorized as (1) shuffling variants: different approaches to shuffle a sequence or subsequence; (2) asymmetry variants: change how symmetrically an amino acid or class of amino acids are positioned in a sequence; (3) property variants: change individual properties (for example, FCR or hydrophobicity) while keeping others constant, while also allowing keeping different aspects of sequence chemistry constant. The sequence optimization approach extends GOOSE's original functionality. Sequence optimization uses the SequenceOptimizer class. This has predefined functionality, but is extensible: any Python function that returns a numerical value can be used with the SequenceOptimizer class. FINCHES-based sequence design is built in through the SequenceOptimizer. Although the sequence optimization approach is more flexible than predefined GOOSE functions, it is slower to generate sequences (typically still on the order of seconds per sequence).

Owing to its speed and flexibility, GOOSE is poised to facilitate the rational design of anything from small numbers of sequences to libraries of thousands for the systematic investigation of sequence–ensemble and sequence–function properties. A key feature of GOOSE is its use of the metapredict (V3) backend to enable rapid, accurate assessment of disorder propensity in designed sequences. Developing a fast and accurate disorder predictor (thousands of seconds per sequence with state-of-the-art accuracy[64]) was essential to enable high-throughput library design.

GOOSE is open source and can be used as a Python library or within a Google Colab notebook. Sequence and sequence variant generation are available online (https://colab.research.google.com/drive/1U9B-TfoNEZbbjhPUG5lrMPS0JL0nDB3o?usp=sharing). Functionality to design IDRs with specific homotypic and/or heterotypic interactions is available online (https://colab.research.google.com/drive/1aJajo1IK66ApFSMwCCBumCZixlfshw7A?usp=sharing). We have provided extensive documentation (https://goose.readthedocs.io/en/latest/index.html), which is not reproduced in the Supplementary Information owing to its length, but can be readily accessed online.

Functionally, GOOSE relies on a stochastic design algorithm that enables it to generate unique sequences even when numerous sequence properties are specified. This is true for both the predefined functions and the SequenceOptimizer class. Generally, fully synthetic sequence generation starts with the creation of a base sequence that comes close to satisfying user-specified input parameters. From here, various functions are used to fine-tune the sequence so that its parameters match the target parameters. For sequence variants, a similar approach is used, with the primary difference that specific constraints are predefined. For some variants, the number of changes from the starting sequence is minimized, ensuring the variant sequence matches the target parameters while remaining as close as possible to the starting sequence. Finally, for sequence generation and variant generation, the sequence is checked for predicted disorder using Metapredict v.3, where the only difference between de novo sequences and variants is that for variants, the disorder scores of the input sequence are used as the minimum values for the generated variant, whereas for fully synthetic sequences, a simple cut-off value is used. GOOSE is in active development, and new features will be added regularly. The version associated with this manuscript is v.0.2.5 at the time of submission.

Sequence property library design rationale

The sequences in this paper that examined the impact of sequence properties on ensemble dimensions were generated by specifying sequence properties within specific ranges, enabling maximal pairwise comparisons in which only one property was varied between pairs of sequences. In particular, sequences were designed with the following quantized sequence properties: NCPR of −0.6, −0.3, 0, +0.3 or +0.6; FCR of 0, 0.3 or 0.6; Kyte–Doolittle hydrophobicity of 1.0 or 3.0 (on a 0–9 scale); and κ (a measure of charge patterning) was set to be between 0.05 and 0.22 (low-to-average, depending on sequence composition) and then above 0.5 for highly clustered sequences. The quantization of hydrophobicity (and 1.0 or 3.0) was selected for two reasons. First, maintaining low hydrophobicity reduces the risk that our synthetic IDRs trigger the unfolded protein response. Second, because hydrophobicity is intrinsically coupled with FCR, enabling independent variation of FCR and hydrophobicity requires lower hydrophobicity scores to accommodate highly charged sequences. Given the scope of the sequence space for 60-residue disordered proteins and the relatively low-throughput experimental characterization used here to ensure high-quality data, we opted to approach our design problem by generating sets of sequence pairs. Each pair enables a specific comparison of a single sequence parameter by holding the others fixed while varying that parameter (for example, net charge or hydrophobicity). By designing our library to multiplex distinct hypotheses, the same sequences could be part of multiple pairs, enabling us to systematically test a collection of hypotheses with relatively few sequences. These sequences were generated using the 'create.sequence' function in GOOSE, which allows generating IDRs while simultaneously specifying length, average hydrophobicity, κ, NCPR and FCR.

Sequences designed by dimensions

To further explore the relationship between basal ensemble dimensions and IDR sensitivity to changes in cellular dimensions, we used GOOSE to generate sequences across a wide range of predicted dimensions. These designs used a simple predefined function in GOOSE called seq_by_re. For these designs, all that we specified was the sequence length and then the objective end-to-end distance in angstroms. Apart from that, the designs were not constrained by any specific sequence properties.

LEA sequence library design

As we did not know which sequence properties would influence the function of each input late embryogenesis abundant (LEA) protein sequence a priori, we used a simple strategy for the LEA library design. Specifically, we used the many-variant functions in GOOSE to design variants of each starting sequence to span a wide range of sequence properties, with some properties altered and others held constant (Supplementary Fig. 5). Each variant was based on a starting LEA sequence. The specific types of variants generated were as follows:

(1) constant_properties variants where FCR, NCPR, hydropathy and κ are held constant in the generated variants; (2) constant_properties_and_class: the same as (1) except that the generated sequence also has the same number of each amino acid by class as the starting sequence; (3) constant_properties_and_class_by_order: the same as (2) except that the order of the amino acids by class is also retained in the generated variant; (4) constant_residues_and_properties: variants where the FCR, NCPR, hydropathy and κ are all held constant and a specified subset of residues (by class) is also held constant in position and number; (5) shuffle_specific_regions: variants where specific subregions of the sequence are shuffled while others are left constant; (6) change_residue_asymmetry: variants where different classes of amino acids were redistributed to be more or less asymmetrical across the sequence; (7) change_hydropathy_constant_class: variants where the average hydrophobicity is altered but the amino acids by class and position are kept constant; (8) change_fcr_minimize_class_changes: variants where the FCR is increased or decreased and changes to amino acids by class are minimized; (9) change_ncpr_constant_class: variants where NCPR is increased or decreased and everything else is held constant; (10) change_kappa: variants where the asymmetry of the distribution of oppositely charged residues is increased or decreased and everything else is held constant; (11) change_dimensions: variants where the end-to-end distance was titrated while changes to the sequence were minimized; (12) FINCHES: variants where the self-to-self ε value as calculated by FINCHES was increased or decreased while minimizing changes to the sequence. For each type of variant, between 6 and 12 sequences were generated for each input sequence. The number of variants per input sequence and per variant type varied slightly because some variants require specific residues to be generated. For example, we did not generate variants that titrated the distribution of prolines across the sequence when the input sequence contained no prolines. After all variants were generated, they were checked for redundancy to ensure that no two sequences were the same.

Benchmarking GOOSE speed

All GOOSE benchmarks in which we measured the speed of sequence generation (Fig. 1c and Extended Data Fig. 1b–d) were conducted on an M1 Max MacBook Pro. All benchmarks were run in a clean Conda environment with Python v.3.13.12. All benchmarking used the GOOSE version associated with v.0.2.5, specifically the git commit 4298c2f. At that time, the following Python package versions relevant to GOOSE performance were used: metapredict v.3.0.1, numpy v.2.4.2, sparrow v.0.2.2 (commit 57324b7) and torch v.2.10.0.

Fluorescence reporter constructs for FRET imaging

Our fluorescence reporter construct places the disordered protein sequences from our library (Extended Data Fig. 2 and Supplementary Table 1) between an N-terminal mTurquoise2 FRET donor and a C-terminal mNeonGreen acceptor[4]. Genes for each IDR were obtained from GenScript and ligated between the two FPs using 5′ SacI and 3′ HindIII restriction sites in a pcDNA3.1(+) backbone.

Mammalian cell culture

U-2 OS cells (ATCC HTB-96, no further validation or mycoplasma testing performed) were cultured in Corning-treated flasks with Dulbecco's modified Eagle's medium (Gibco Advanced DMEM:F12 1×) supplemented with a final concentration of 10% FBS (Gibco) and 2.5 mM L-glutamine (Gibco). For live-cell microscopy experiments, 8,000 cells were plated in each well of a µ-Plate 96 Well Black treated imaging plate (Ibidi) and allowed to adhere overnight (around 16 h) before transfection. Cells were incubated at 37 °C and 5% CO_2. Before transfection, the medium was replaced with new, warmed medium. XtremeGene HP (Sigma-Aldrich) was used to transfect FRET construct plasmids into U-2 OS cells according to the manufacturer's protocol. Cells were then incubated at 37 °C and 5% CO_2 for 24 h.

Live-cell microscopy for FRET imaging

Images were taken on the Olympus IX83 microscope equipped with a laser-guided autofocus system and using a ×20/0.8 NA dry objective. Excitation light was produced with a Spectra X light engine (Lumencor), and data were collected on two linked ORCA-fusion Gen-III sCMOS cameras (Hamamatsu Photonics). For donor, acceptor and direct acceptor channels, cells were imaged at room temperature before and after perturbation with 5 ms exposure times and 2.5% LED intensity. Imaging was performed by exciting mTurquoise2 at 438 nm (438/29) for donor and acceptor channels, mNeonGreen at 511 nm (511/16) for the direct acceptor channel, and Hoechst dye at 390 nm (390/22) for the nucleus. The emitted light was passed downstream through a triple-bandpass dichroic (467/24, 555/25, 687/145), then split and directed to the two linked cameras using a 510 nm downstream dichroic housed in a Gemini Optical Splitter (Hamamatsu Photonics). Light additionally went through post-dichroic filters also housed in the Gemini Optical Splitter before reaching the cameras, with light directed towards camera 1 (longer wavelength) passing through a 544/24 emission filter and shorter wavelength light (camera 2) passing through a 482/25 emission filter.

Cells were stained with Hoechst 33342 (Thermo Fisher Scientific) to visualize nuclei by adding Hoechst stock (20 mM) to PBS (Gibco) at a ratio of 0.25 µl to 1.5 ml. Shortly before imaging, 250 µl of this solution was added to each well and incubated for 5 min, then washed with fluorobrite, after which a final volume of 250 µl Fluorobrite (Gibco) was added. The imaging process was started around 15 min after the final addition of Fluorobrite.

Osmotic perturbation imaging for FRET imaging

Osmotic perturbation solution of 1.8 M mannitol was prepared by dissolving 9.84 g of mannitol (Acros Organics) in a final volume of 30 ml Fluorobrite (Gibco) shortly before imaging. After imaging in isosmotic fluorobrite medium, 50 µl of the mannitol solution was added to four adjacent wells using a multichannel pipette to achieve a +250 mOsm challenge (550 mOsm total). To ensure mixing, the solution was pipetted up and down ten times, after which the four wells were imaged. This process was then repeated for all wells in the plate. Imaging for each group was completed between 45 s and 2 min after perturbation, depending on the well in the imaging order.

Image analysis for FRET imaging

Images were analysed using a custom Python script (Supplementary Fig. 8) available on the associated GitHub repository. Cell masks were created by segmenting using CellPose[65] with default settings on the FRET donor channel. A nucleus mask was created by thresholding using Otsu's method on the Hoechst channel. Image sets before and after perturbation were registered to each other (without the Hoechst channel) using a rigid registration algorithm[66]. A constant value for each channel, determined from the mean values of empty wells, was then subtracted (except for the Hoechst channel) from all pixel values to remove background effects at low fluorescence intensities from the ratiometric FRET efficiency metrics. Areas where background subtraction caused signal values to approach 0 are removed from the masks and are not counted in the calculated metrics. Hoechst masks are labelled to their corresponding overall cell mask, and small holes in the Hoechst mask are filled in.

From these finalized masks, a variety of metrics are calculated for each cell. These include mean intensities for the FRET donor, FRET acceptor and direct acceptor in the whole cell, nucleus and cytoplasm. Cell FRET efficiencies before and after perturbation ($E_{f,basal}$ and $E_{f,perturbed}$, respectively) were calculated by $E_f = F_A/(F_A + F_D)$. Here, F_D is the mean intensity of the donor, and F_A is the mean intensity of the acceptor after fluorescence bleedthrough and cross-excitation corrections (Supplementary Fig. 9). $E_{f,perturbed}$ channels were additionally corrected to

account for total fluorescence intensity changes observed after perturbation (Supplementary Fig. 10).

The FRET efficiency change owing to osmotic perturbation was calculated as $\Delta E_f = E_{f,perturbed} - E_{f,basal}$. Data are then filtered to remove dead cells, strange morphologies, missing in after-images and so on. All filters used are displayed per construct (Supplementary Fig. 11) and include cell area, nucleus area, cell circularity, ratio between nucleus and cell areas, direct acceptor maximum, and direct acceptor intensity ratio after and before perturbation. Cells are then further filtered based on the mean intensity of their direct acceptor channel to avoid intermolecular FRET effects (Supplementary Fig. 12).

Statistical analysis for FRET imaging

The statistical analysis of all experimental data was performed using the SciPy library in Python. Experiments were performed in 96-well plates across multiple cell passages and days, with each well plated and transfected individually. We therefore treated each well as a biological replicate of the experiment. Thus, the median values for all cells measured per well were used to generate a single violin plot. We excluded wells that contained fewer than 25 cells. The s.d. and average values were calculated from the medians of all wells from each experimental condition. To assess the significance of the differences between two constructs, a two-sided Student's *t*-test was performed between all medians of the two constructs.

Bioinformatic analysis of FRET construct sequences

Bioinformatic sequence analysis for metrics such as NCPR, FCR and κ was performed using localCIDER and sparrow (https://github.com/idptools/sparrow/). Disorder prediction was performed using metapredict (V2-FF).

Coarse-grained simulations of FRET library sequences

Molecular dynamics simulations were performed using Gromacs (v.2021.5)[67] and the Martini3-IDP_SC force field[68]. IDRs were coarse-grained and placed in a cubic box using Polyply[69] and solvated using the Insane Python script with a NaCl concentration of 0.1 M. The system was minimized using steepest descent for 5,000 steps, with flexible bonds and a force convergence tolerance of 100 kJ mol^{-1} nm^{-1}. The system then underwent NVT relaxation for 1 ns with a timestep of 10 fs using the velocity-rescaling thermostat, followed by an NPT relaxation for 1 ns with a timestep of 10 fs using the velocity-rescaling thermostat and Berendsen barostat. The system was then equilibrated for 100 ns with a 20 fs time step using the velocity-rescaling thermostat and the Berendsen barostat. A 2 µs production run with a timestep of 20 fs was then performed using the velocity-rescaling thermostat and Parrinello–Rahman barostat. The first 250 ns were discarded to account for equilibration. The cut-offs for van der Waals and Coulombic interactions were set to 1.1 nm, with a dielectric constant of 15. The temperature was set to 30 °C, the pressure to 1 bar and frames were saved every 500 ps. Simulations were run in triplicate from different starting points, and ensemble convergence was quantified using the Hellinger distance on the radius of gyration distributions, with a convergence threshold of <0.3. The reported values and errors are averages and s.d. from the three triplicates.

Bacterial expression and purification of FRET constructs

For in vitro expression and purification, FRET constructs were cloned into a pET-28a(+)-TEV backbone. BL21(DE3) *Escherichia coli* cells (New England Biolabs) were transformed with these plasmids according to the manufacturer's protocol and grown in lysogeny broth (LB) medium with 50 µg ml^{-1} kanamycin. The cultures were incubated at 37 °C while shaking at 250 rpm until an optical density at 600 nm (OD_{600}) of 0.65 was reached (approximately 5 h), then induced with 1 mM isopropyl β-D-1-thiogalactopyranoside and incubated for 18 h at 20 °C while shaking at 250 rpm. Cells were collected by centrifugation for 30 min at 3,000*g*, and the supernatant was discarded. The remaining pellet was lysed in lysis buffer (50 mM NaH_2PO_4, pH 8.0, 0.5 M NaCl) using a QSonica Q700 Sonicator (QSonica). The lysate was centrifuged for 30 mins at 21,100*g*, the supernatant was collected and Ni-NTA agarose beads (Thermo Fisher Scientific) were added to the lysates. After incubation with lysates at 4 °C for 15 min, the beads were centrifuged at 18,000*g* for 15 min, and the flow-through was discarded. Ni-NTA beads were washed twice with wash buffer (50 mM NaH_2PO_4, pH 8.0, 0.5 M NaCl, 20 mM imidazole). The FRET construct was eluted with 50 mM NaH_2PO_4, pH 8, 0.5 M NaCl and 250 mM imidazole, and buffer-exchanged into protein buffer (20 mM sodium phosphate buffer, pH 7.4, 100 mM NaCl). Constructs were further purified using Amicon Ultra 0.5 ml Centrifugal Filters (MilliporeSigma), with a 50 kDa cut-off filter. The purified FRET constructs were divided into 100 µl aliquots, flash-frozen in liquid nitrogen, and stored at −80 °C in protein buffer. The protein concentration was measured after thawing and before use using UV–Vis absorbance at 506 nm (the peak absorbance wavelength of mNeonGreen). Protein purity was assessed by SDS–PAGE after thawing and before use. To verify the integrity of the FRET construct and fluorophore stoichiometry, we measured the absorbance spectrum of each sample before each FRET assay. We report only samples that displayed an absorbance ratio 506/434 (peak mNeonGreen absorbance/peak mTurquoise2 absorbance) of 2.5 ± 0.5, a reasonable ratio given the difference in the molar extinction coefficients of mTurquoise2 and mNeonGreen (34,000 l mol^{-1} cm^{-1} and 116,000 l mol^{-1} cm^{-1}).

In vitro FRET experiments

In vitro FRET experiments were conducted in black, flat-bottom 96-well plates (Thermo Fisher Scientific) using a CLARIOstar plate reader (BMG Labtech). Protein buffer and purified protein solution were mixed in each well to a final volume of 150 µl, containing the buffer and the FRET construct at 1 µM protein. Fluorescence measurements were taken from the top of the plate, at a focal height of 6.8 mm, with gain fixed at 1,200 for all samples. For each FRET construct, three biological replicates were obtained from different expressions, with 3 technical replicates per expression performed. Fluorescence spectra were obtained for each FRET construct by exciting the sample in a 16-nm band centred at λ = 420 nm, with a dichroic at λ = 436.5 nm, and measuring fluorescence emission from λ = 450 to 600 nm, averaging over a 10-nm window moved at intervals of 0.5 nm. Base donor and acceptor spectra were obtained using the same excitation and emission parameters on solutions containing 1 µM mTurquoise2 or mNeonGreen alone.

Calculation of in vitro FRET efficiencies

The E_f of each FRET construct in each solution condition was calculated as described previously[4]. In brief, this was done by linear regression of the fluorescence spectrum of the FRET construct against the separate donor and acceptor emission spectra under the same solution conditions (to correct for solute-dependent effects on fluorophore emission). The E_f was calculated using the following equation:

$$E_f = 1 - \frac{F_d}{\frac{Q_d f_d}{Q_a f_a} F_s + F_d}$$

where F_d is the decoupled donor fluorescence, F_s is the decoupled acceptor fluorescence, f_d is the area-normalized donor spectrum, f_a is the area-normalized acceptor spectrum, Q_d = 0.93 is the quantum yield of mTurquoise2 and Q_a = 0.8 is the quantum yield of mNeonGreen[70].

Yeast desiccation library

The sequence designs were reverse-translated into DNA sequences. To ensure proper sequence coverage and distribution across our sequence library, we developed a simple Python package that allows for simultaneous optimization of codon usage and GC content. With this, we were able to generate DNA sequences optimized for

S. cerevisiae codon usage, with an average GC content of around 40% and a GC s.d. of 0.76% across the library. This tool is publicly available at GitHub (https://github.com/ryanemenecker/kea). The resulting DNA sequences were synthesized on-chip (Twist Bioscience) and cloned into a low-copy number plasmid with a strong promoter using 15-bp homology regions at the 5′ and 3′ ends through in-cell homologous recombination[71]. The DNA library was obtained from Twist Biosciences. Each DNA sequence included 12 nbp of homology to the plasmid backbone at both the 5′ and 3′ ends (5′-ACTAGTGGATCC…CTCGAGTCATGT-3′), and amplified in parallel 16 times in 25 µl reactions with NEB Q5 polymerase in 10 cycles to extend the homology to the backbone (forward primer: 5′-TCTAGAACTAGTGGATC-3′; reverse primer: 5′-ACTAATTACATGACTCGAG-3′). The library was cleaned up using a 1.8× ratio of Magmax beads to PCR product, and the clean DNA was quantified with a Qubit 2.0 dsDNA Broad Range assay, and specific amplification was confirmed on a 3% agarose gel. The backbone plasmid p413GPD-hTPI was purchased from Addgene and was linearized for cloning by amplification with NEB Q5, after a preliminary digestion with BamHI-HF (NEB). It was amplified in 24 cycles (forward primer: 5′-CACTAGTTCTAGAATCCGTCG-3′; reverse primer: 5′-TCGAGTCATGTAATTAGTTATGTC-3′). The amplified product was cleaned with a Monarch NEB PCR and DNA Cleanup kit, then digested with FastDigest DpnI from Thermo Fisher Scientific in Green Buffer for direct loading onto a 1% agarose gel. The amplified product was excised from the gel using the Monarch Gel Extraction kit.

To ensure single or low plasmid copies in yeast cells and high expression levels, the p413GPD backbone contained a CEN6–ARS4 integration site and a GPD promoter. Transformation was carried out in yeast by electroporation[72], and the copies of double-stranded DNA (dsDNA) of library:backbone were in a 5:1 ratio to ensure an abundance of insert. Lower library:backbone ratios resulted in fewer transformants. In total, 18 transformations were conducted in parallel to guarantee there were enough transformants for the library size, 10× coverage. To achieve this, we plated 200 histidine-deficient agar plates, each with approximately 400 colonies and scraped a total of around 80,000 colonies into a 20% glycerol stock.

The plasmid library was transformed into BY4742 yeast cells, and the resulting library was used for experiments, with at least 6×10^6 cells to ensure proper coverage. For each experiment, the cells were grown in histidine-deficient synthetic complete (SC-His) medium to select for library expression. We ensured that cells did not experience any stress before desiccation by maintaining the culture in log phase using turbidostats. This prevents the yeast from biochemical priming, which confers natural desiccation protection[73] but would obfuscate the protective effect of the IDRs. The yeast was then subjected to desiccation by pipetting them on nitrocellulose filter paper for 8 h in a humidity-controlled chamber with low relative humidity (<10%). After desiccation, each filter paper was rehydrated in SC-His under shaking, and the yeast culture extracted from the filter paper was allowed to grow in a turbidostat, ensuring the culture remains in logarithmic growth for 24 h (Extended Data Fig. 7b). This regrowth guarantees that cells that did not survive desiccation are depleted, while surviving cells proliferate. The experiment was run in at least triplicate for each experimental repeat, and samples were collected from each replicate before and after dessication–rehydration. Plasmids were extracted from each sample, the variable region of the sequence was amplified and the amplicons were prepared for sequencing. As a control, we compared the pooled library culture to a yeast culture containing an empty (non-expressing) vector under the same promoter and observed significantly faster growth rates in the library culture than in the EV culture (Extended Data Fig. 7b).

Yeast desiccation and rehydration

Yeast cells grow in distinct phases: lag, log and stationary. In stationary phase, yeast cells are stressed and compete for resources, so they prime themselves and show increased viability under desiccation and other stressors, such as heat shock. As we want to profile the extent of protection from the IDR library, we desiccate at mid-log phase, when they are unprimed. Yeast cells transformed with the library were thawed from a glycerol stock for 30 min on ice. A starter culture of approximately 5×10^6 cells per ml was incubated for 5 h to reach mid-log phase (1.5×10^7 cells per ml). Cells were then spun down at 6,000*g* for 6 min, washed twice with sterile Milli-Q H_2O and resuspended in one-tenth the volume of water. Then, 300 µl of mid-log cells was aliquoted onto filter paper and dried in a desiccator cabinet containing a saturated salt solution, NaOH, which maintains a relative humidity of 9%[74] for 8 h. The dried cells were taken from the desiccation cabinet and resuspended in 15 ml of selective medium. To ensure that dried cells were lifted from the filter paper, the cells and filter paper were incubated in SC-His media in a flask for 1 h. Once lifted, a starter culture of 0.2×10^7 cells per ml was inoculated in the turbidostat (purchased from Pioreactor) for outgrowth. The OD_{600} was calibrated with a cell counter (DeNovix CellDrop FLi), and growth was monitored once every 4 s. Yeast cells recovering from desiccation were incubated in 15 ml of SC-His medium for approximately 24 h to keep the samples at mid-log phase before extracting the plasmid library. The turbidostat was programmed to remove 2 ml of cells and add 2 ml of fresh medium to dilute the culture when it reached a density of 2×10^7 cells per ml. Stirring was used in place of shaking at 500 rpm, and cells were incubated at 30 °C.

Viability by regrowth rates

Regrowth rates post-desiccation were determined from OD readings obtained in the Pioreactors. A starter inoculation of 0.2×10^7 cells per ml was allowed to recover for 24 h. The cell density was measured every 4 min (Extended Data Fig. 7b). Pioreactor settings were as follows: stirring at 500 rpm, 30 °C, removal of 2 ml cells and addition of fresh 2 ml selective medium (SC-His) at cell counts of 2×10^7 cells per ml. The OD was calibrated to cell concentration using a cell counter (DeNovix CellDrop FLi). The library was recovered in triplicates, and an EV control was used as the baseline recovery after desiccation.

Illumina sequencing

Plasmids were extracted using the Spin Miniprep Kit (Qiagen) from yeast cells before and after desiccation–rehydration according to a modified glass-bead lysis protocol. The extracted plasmid library was amplified in 24 cycles and purified with Ampure XP magnetic beads. DNA fragments larger than around 1 kb were removed using a two-step SPRI bead cleanup (AMPure XP, Beckman Coulter). In brief, PCR products were incubated with 0.5× bead volume to selectively bind to high-molecular-mass fragments, and the supernatant containing smaller fragments was transferred to a new tube. DNA was then recovered from the supernatant by the addition of 0.8× bead volume, followed by two washes with 70% ethanol and elution in 20 µl nuclease-free water. DNA was quantified on the Qubit 2.0 system using a dsDNA high-sensitivity kit. Then, 20 ng of DNA was used for downstream adaptor ligation and barcode amplification using the NEBNext library preparation kit according to the manufacturer's protocol without size selection. The DNA was quantified on the Qubit fluorometer, and the fragment size distribution was checked on an Agilent TapeStation 4200 using the D1000 High Sensitivity ScreenTape assay. The samples were sent to Azenta for sample preparation and sequencing on their MiSeq platform.

Illumina sequencing analysis

For each experimental repeat, at least three biological samples were sent for sequencing. Reads were aligned using Bowtie 2 and normalized using DESeq2 (ref. 75). FastQC files were obtained from Azenta and aligned with Bowtie 2 (ref. 76) to the reference library. Samtools[77,78] indexed and sorted the aligned library, and idxstats produced the aligned-reads output. DESeq2 differential expression analysis normalized reads between libraries to quantify the FC between the number

of reads before and after desiccation, along with *P* values adjusted using Benjamini–Hochberg correction to control for the FDR. The weighted $\log_2$[FC] average was calculated as $\underline{x} = \frac{\sum_i w_i \times x_i}{\sum_i w_i}$ where $w_i = \frac{1}{\text{s.e.}_i^2}$ and the pooled s.e. was calculated as $\text{s.e.}_{\text{pooled}} = \frac{1}{\sqrt{\sum_i w_i}}$.

Library sequence property analysis

In total, 2,300 sequences were mapped across 3 repeats. In total, there were 11 template variants, chosen because they had predicted protection ranges from a small preliminary library. At a minimum, 107 variants are generated, up to 282. To plot the correlations between sequence features and variants, a filter of ±7% of the wild-type variant was applied to exclude redundant datapoints and denoise the scatterplot, as many sequences have identical or highly similar sequence feature values to the wild type, per the library design. Sequence features were calculated using CIDER[11], sparrow[37] and NARDINI+[18] (Supplementary Table 4).

Generation of yeast strains for use in self-assembly and client recruitment assays

Yeast strains generated for IDR self-assembly and client recruitment assays were created using a modified lithium acetate–single-stranded DNA (ssDNA)–PEG transformation protocol[79]. In brief, BY4741 was streaked from glycerol and grown for 2 days. Colonies were selected and grown in YPD overnight at 30 °C, with shaking at 190 rpm. The next morning, 10 ml of yeast culture was centrifuged, and the yeast cells were resuspended in 2 ml of lithium acetate buffer (0.1 M lithium acetate, 10 mM Tris-HCl pH 8.0, 1 mM EDTA) and then centrifuged again. The yeast cells were then resuspended in 150 µl of lithium acetate buffer. Next, 100 µl of the resuspended cells was added to a 1.7 ml sterile microcentrifuge tube containing 10 µl of 10 mg ml^{-1} ssDNA and 10 µl of plasmid DNA in 10 mM Tris (50–500 ng µl^{-1} DNA). Next, 280 µl of PEG solution (0.5 g ml^{-1} PEG 3350, 0.1 M lithium acetate, 10 mM Tris-HCl pH 8.0, 1 mM EDTA) was added to the tube, and the tube was then vortexed. After incubation for 30 min at room temperature, 43 µl of DMSO was added to the tube. After vortexing, the yeast cells were heat-shocked at 42 °C for 15 min and then placed on ice for 1 min. Next, the yeast cells were pelleted and the supernatant was removed. The yeast cells were then resuspended in 500 µl of the appropriate selective synthetic dropout (SD −His, SD −Ura, or SD −His −Ura), centrifuged again and resuspended in 200 µl of the appropriate selective synthetic dropout. Yeast cells were then plated onto the appropriate selective synthetic dropout and grown at 30 °C.

Yeast microscopy for IDRs designed to form assemblies

For all yeast microscopy examining assembly formation, the base strain was BY4741. Clients and scaffolds were expressed under the TDH3 promoter from an ARS4–CEN6 plasmid containing a URA3 or HIS3 auxotrophic marker. For microscopy experiments, the required strain was streaked onto the appropriate selective plate and incubated at 30 °C to obtain single colonies. The day before imaging, a single colony of each required strain was selected and grown in six-well plates in 2 ml liquid cultures at 30 °C and 185 rpm. The next morning, around 3 h before imaging, each strain was subcultured at a 1:1 ratio to maintain log-phase growth. Then, 1 h before imaging, slides were prepped. Imaging was performed using the Ibidi µ-Slide 18-well glass-bottom 1.5H coverslip (Ibidi, 81817; Thermo Fisher Scientific, NC1752257). The 18-well µ-Slide was treated with 40 µl of 1 mg ml^{-1} concanavalin A (Thermo Fisher Scientific, J61221-MD) for 15 min to help yeast adhere to the slide and remain stationary during imaging. Next, 1 ml of each culture was spun down and resuspended in 100 µl of the remaining supernatant. The concanavalin A solution was then aspirated from each well. Each well was then washed with 40 µl of water. Next, the water was aspirated, 40 µl of culture was added to the well and the well was left to incubate for 5 min. The non-adherent cells were then aspirated, and 40 µl of fresh medium was added to the well. Next, the yeast cells were immediately taken for imaging. Imaging was carried out at the Washington University in St Louis Center for Cellular Imaging (WUCCI) using the Zeiss LSM 880 Confocal with Airyscan microscope equipped with a Plan-Apochromat ×100/1.40 NA Oil DIC M27 objective. The mTagBFP2 imaging used a 405 nm diode laser at 1%, with the PMT detector gain set to 720, a detection wavelength of 410–530 nm and around 1.0 Airy units. The mScarlet3 imaging used a 561 nm DPSS laser at 0.75%, a PMT detector gain of 720, a detection wavelength of 582–754 nm and around 0.7 Airy units. The mTurquoise2 imaging used the 458 nm argon laser at 1.0%, PMT detector gain set to 750, detection wavelength from 462–580 nM and around 0.99 Airy units.

Approach and rationale for scaffold and client designs

There are many potential solutions to the design objective of generating sequences that do or do not self-assemble and sequences that do or do not get recruited to assemblies. When we first designed sequences to test whether we could use FINCHES to create IDRs that self-assemble, we chose to test a range of ε values from highly attractive to highly repulsive to see how well ε correlates with self-assembly in cells (Fig. 4e). We designed these IDRs to be 200 amino acids, reasoning that 200 amino acids should be sufficiently long to provide the multivalency necessary for self-assembly. When we designed our scaffold IDRs (Fig. 4f), we increased their length to 400 amino acids to ensure sufficient interaction sites for homotypic interactions necessary for self-assembly and heterotypic interactions necessary to recruit client sequences. We decided to use a minimum length of 200 amino acids for the clients, reasoning that using a relatively long IDR for our client would increase our specificity for the target scaffold and therefore improve our odds of successfully designing a client that partitions into a given target assembly. However, we note that these numbers are arbitrary, and one could probably design assemblies with sequences much shorter than 400 amino acids that are still capable of recruiting specific clients. The objective of these experiments was not to explore the vast number of potential solutions to this design problem, but rather to demonstrate that it is possible to do so using GOOSE. In future studies, it would be interesting to systematically titrate the lengths of the client and scaffold sequences, or the homotypic interaction strength of the client, to examine how these factors affect system behaviour; however, this is outside the scope of the current study.

Sequences designed to self-assemble or target assemblies were generated using the SequenceOptimizer in GOOSE. The MeanSelfEpsilon property with the model set to 'mpipi' was used to modulate whether a sequence was self-attractive or self-repulsive by specifying a negative or positive target ε value, respectively. For generating client sequences that target (or do not target) self-assemblies, the MeanEpsilonWithTarget property with the model set to 'mpipi' was used, where the target_value was set to be negative (attractive) and the target_sequence was set to the sequence of the assembly (scaffold 1 or scaffold 2). Moreover, the Complexity property was added when generating sequences using GOOSE's SequenceOptimizer with a target_value of 1 to avoid highly repetitive sequences that would be difficult to synthesize. Finally, the FractionDisorder property with a target_value of 1 was applied to all sequences generated by the SequenceOptimizer. The error tolerance from objective target values was set to 0.001, and max_iterations was set to 10,000.

Quantifying partition coefficients for colocalization analysis

Quantification of partition coefficients for colocalization analysis of the GOOSE-designed IDRs with the synthetic self-assemblies was performed with a custom Python script that used the scikit-image and SciPy libraries. The cells were imaged as *z* stacks, and a substantial number of out-of-focus planes were in the data. To avoid data accuracy issues caused by out-of-focus cells, the first and last *z* planes of the *z* stack were discarded, leaving only the middle 50% for analysis.

First, cells were segmented using the client channel, slice by slice. The global background was then determined using the median absolute deviation of the image array. The foreground was segmented using Li's Minimum Cross Entropy algorithm, which thresholds images with heavily skewed object populations (such as self-assemblies)[80]. Binary masks were further refined using median filtering, morphological closing and hole filling. Touching cells were separated using a Euclidean distance transform coupled with a watershed algorithm, and then further filtered by size exclusion to remove noise and truncated cells. Finally, the cell boundary was uniformly shrunk by 10 pixels to avoid including the area outside the cell. This was necessary to avoid skewing the average intensity calculated for the client signal that did not overlap with the assemblies (which would artificially reduce the average client signal and thus increase the calculated partition coefficients). Next, assemblies were segmented independently using the scaffold channel. This used a two-pass statistical approach: first, the cellular foreground was identified using a 50× median absolute deviation threshold, and then refined using median filtering. This was followed by morphological opening to remove noise, hole filling and separation of individual assemblies using a local maxima watershed. Assemblies with fewer than 50 pixels or with intensity values below 2,000 intensity units (values empirically determined) were then filtered out.

Once the cell boundaries and assembly boundaries were identified for every z plane, a '2.5-dimensional axial linking approach' was used. In this approach, assembly masks generated on individual two-dimensional slices were evaluated for overlap across the z axis. Overlapping assemblies were assigned a single identifier (assuming that assemblies in the same location across multiple z planes were the same). For assemblies spanning multiple z planes, each plane was examined, and the plane with the highest intensity was used to assess the client's colocalization with that assembly. All other z planes were discarded, such that each assembly was quantified only once. The average fluorescence intensity was measured for the client protein inside the assembly mask (I_{in}) and in the surrounding area (I_{out}). Moreover, the background fluorescence (I_{bg}) was estimated from non-cell regions and subtracted from the total fluorescence (I). The partition coefficient was calculated as $\frac{I_{in} - I_{bg}}{I_{out} - I_{bg}}$. For quantification, Li's minimum cross-entropy threshold was applied to eliminate cells expressing the client at too low a level. As a final quality control step, we set up our analysis to output a .png file showing the raw channels for the assembly and the client, next to what was used as the mask for 'inside the assembly' and 'outside the assembly', such that we could examine each datapoint individually and ensure there were no abnormalities with the quantification. For example, if an assembly was localized to the nucleus for one of the constructs that tended to have strong nuclear client signal, that datapoint would be excluded because it would measure the overlap in assembly signal to strong client nuclear signal, which would suggest strong colocalization signal that is largely owing to the nuclear signal of the client rather than true colocalization with the scaffold.

To assess whether individual constructs differ significantly in their partitioning behaviour, we performed pairwise Mann–Whitney U-tests (two-sided) on all unique pairs of constructs within each scaffold condition. The Mann–Whitney U-test was chosen as a nonparametric alternative that makes no assumptions about the underlying distribution of partition coefficients. For each scaffold, given k constructs, we computed $k/2$ pairwise comparisons. To control the family-wise error rate across the multiple comparisons within each scaffold, P values were corrected using the Holm–Bonferroni step-down procedure, which provides strong control of the family-wise error rate while being uniformly more powerful than the classical Bonferroni correction. Corrected P values were considered significant at $\alpha = 0.05$, with additional thresholds at 0.01 and 0.001 for graphical annotation. All statistical tests were performed using SciPy (scipy.stats.mannwhitneyu), and multiple-testing correction was applied using statsmodels (statsmodels.stats.multitest.multipletests, method 'holm')[81].

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Data and analysis scripts used for figures and analysis in this paper are available at GitHub (https://github.com/sukeniklab/GOOSE_2026 and https://github.com/holehouse-lab/supportingdata/tree/master/2026/GOOSE_2026). Raw sequencing data have been deposited at Zenodo[82] (https://zenodo.org/records/18774462).

Code availability

GOOSE is fully open source and is available at GitHub (https://github.com/idptools/goose/). Documentation for using GOOSE is available online (https://goose.readthedocs.io/en/latest/). A subset of GOOSE's functionality is available across two Colab Notebooks. For sequence generation and variant design, use the following notebook: https://colab.research.google.com/drive/1U9B-TfoNEZbbjhPUG5lrMPS0JL0nDB3o?usp=sharing. For creating sequences with specific homotypic and/or heterotypic IDR–IDR interactions, use the following notebook: https://colab.research.google.com/drive/1aJajo1IK66ApFSMwCCBumCZixlfshw7A?usp=sharing.

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Author contributions R.J.E., A.S.H. and S.S. conceptualized the study. R.J.E., A.S.H. and S.S. designed the methodology. R.J.E., A.S.H. and J.M.L. contributed to software development. K.H., T.B., K.G., R.J.E., N.M.S., B.N., K.C.K., J.K.N. and L.M.D. generated experimental data. T.B., K.G., N.M.S. and A.M. conducted in-cell FRET experiments. K.C.K. performed in vitro FRET experiments. J.K.N. conducted molecular dynamics simulations. R.J.E. performed yeast self-assembly experiments. K.H., B.N. and L.M.D. generated data for the yeast desiccation library. A.S.H., S.S., R.J.E., T.B., K.G. and K.H. carried out data analysis. A.S.H., S.S., J.M.L. and K.H. provided resources and acquired funding. K.H., T.B., K.G., A.S.H., S.S. and R.J.E. performed visualization and wrote the manuscript. R.J.E. formatted the Extended Data and Supplementary Information.

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Additional information

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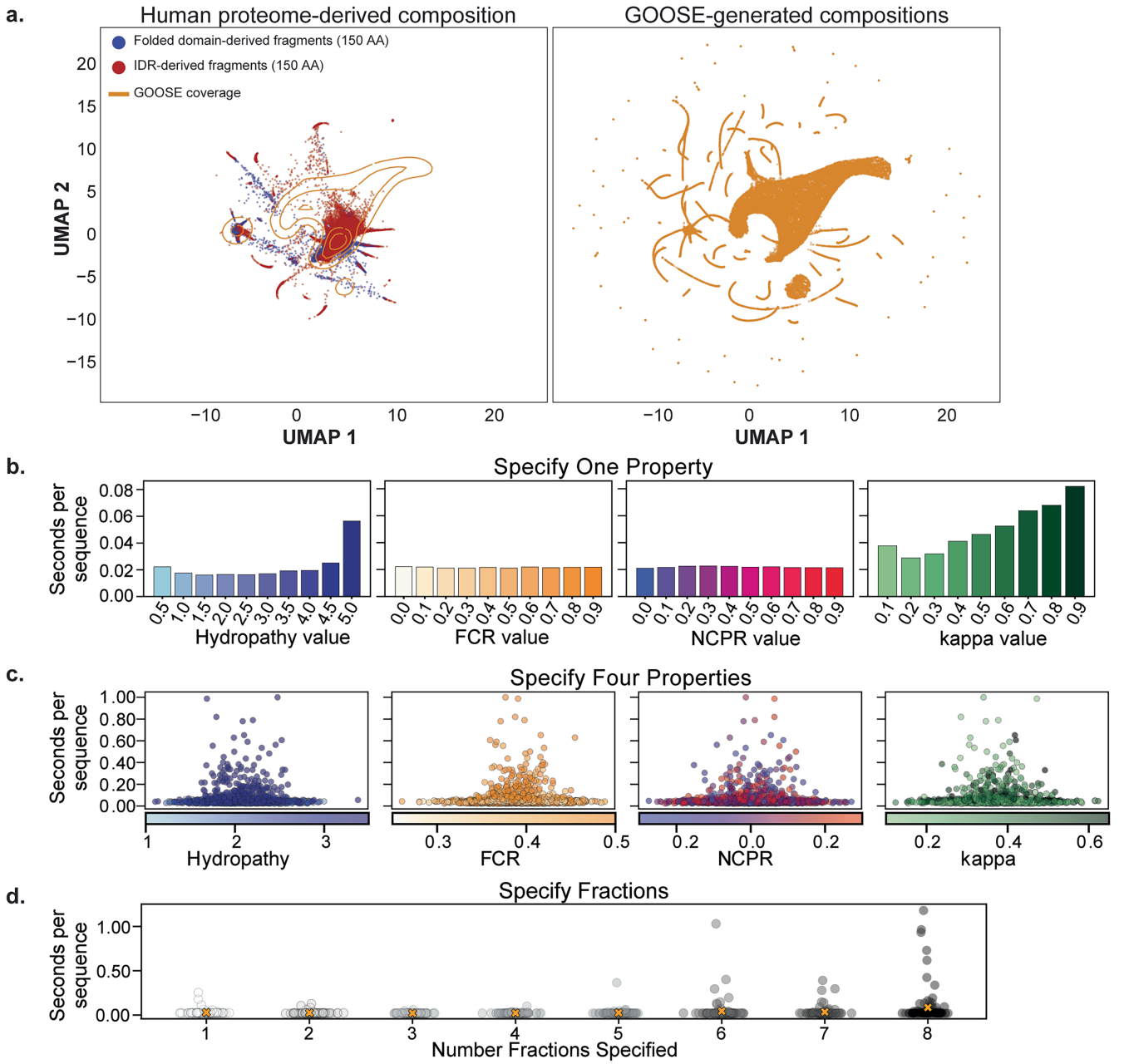
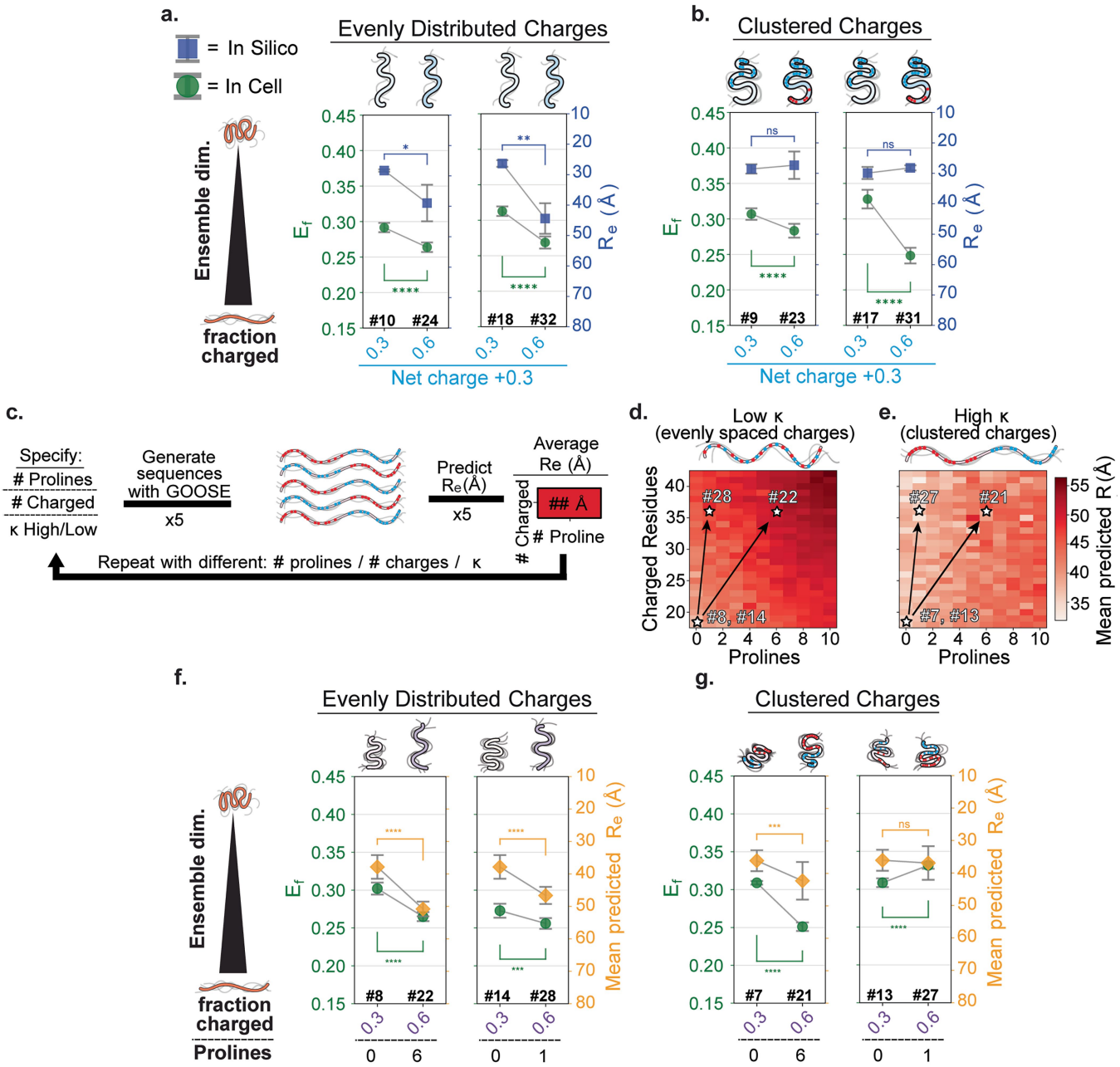


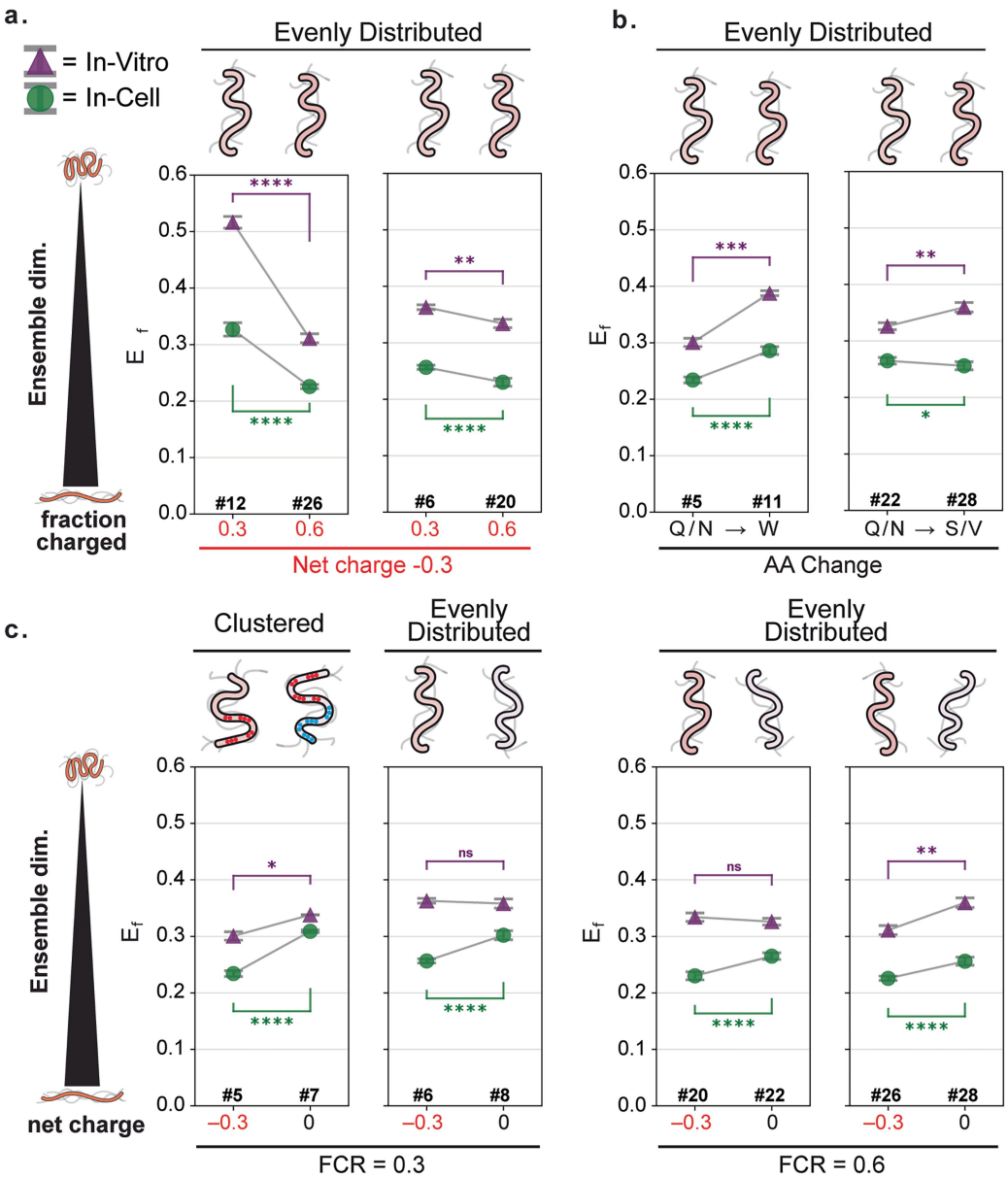
Extended Data Fig. 1 | GOOSE can rapidly generate sequences and sample sequence space outside of naturally occurring IDRs. a. Uniform Manifold Projection (UMAP) showing projected compositional space occupied by 150-amino acid fragments from the human proteome (left) or generated by GOOSE (right). UMAP was generated by embedding sequences into an L = 20 vector, where each element corresponds to the fraction of each amino acid. Human proteome fragments were identified by segmenting the human proteome into disordered regions and folded domains, then taking 150-amino-acid subfragments within those regions (step size of 50). This yielded 33,626 IDR fragments and 85,098 folded fragments. For GOOSE, we generated a minimum of 10 sequences by titrating across amino acid fractions, properties (NCPR, FCR, hydropathy, kappa), and amino acid classes (aromatic, aliphatic, polar, positive, negative, glycine, proline, cysteine, histidine), titrating from minimum to maximum possible values (step size 0.1). This yielded 67,427 150-residue GOOSE-derived segments. **b**. Time to generate sequences where only a single property is specified. Times are averaged over 200 generated sequences. **c**. Time to generate sequences where four properties are specified. Each point represents the time required to generate a single sequence. Each plot shows the property for that sequence. A total of 1,656 generated sequences are shown as they titrate across hydrophobicity, FCR, NCPR, and kappa values. **d**. Time to generate sequences where different numbers of amino acid fractions are specified. A total of 100 sequences were generated for each number of fractions specified. Each point is an individually generated sequence. The orange 'X' shows the average value across the 100 sequences. For **b**.-**d**., all sequences are 200 amino acids in length.

#	Sequence	FCR	NCPR	κ	H	Dis.	Self ε
1	QNNNQQQNQQNQNNQNNNQNNNNQNQNNQQQQNQQQNNNQQQNNQQNNNNQQQNNNQNNN	0	0	-1	1	0.88	8.19
2	THNHHSTPGTPGHHHPGSPHSPHPTHTTPSHHGTGGGHGGSTTQSHSNGSATGQHGSSGP	0	0	-1	3	0.89	3.22
3	THNHHSTGTGHHHGSHSHTHTTSHHGTGGGHGGSTTQSHSNGSTGQHGSSGPPPPPPPAP	0	0	-1	3	0.91	3.22
4	THNHHGPSTGTGPHHHGSPHSHTHPTTSHHPGTGGGPHGGSTPTQSHSANGSTGPQHGSS	0	0	-1	3	0.9	3.22
5	HNNQQQQQNNQNNQNQQNNQNNNNNQQQQNQQEDEQQDDENQDDDQQDEDQNEDEQQDDE	0.3	-0.3	-1	1	0.94	32.12
6	EDDEHNNQDQQQQENNQNDNQNQDQNNQDNNNNNQQQEQNQQDQQNQDQQQNDQQEEDED	0.3	-0.3	-1	1	0.92	32.12
7	KKKRSGQNRNQNNRQQQQKNQNQRQNQNKNNQQNQNNENNNNDNNQQENNQQENQDDDDD	0.3	0	0.53	1	0.92	8.36
8	REKRSGQNKNQNNRQQQQDNQNQEQNQNDNNQQNQNNDNNNNKNNQQDNNQQRNQEKDKD	0.3	0	0.13	1	0.95	13.28
9	KKRSGKRRHHKKRHQKRRQHRKKQQRKKQNQNNNNNQNQQQQHHNQQNNQNNPNNNNHHP	0.3	+0.3	-1	1	0.85	16.45
10	KRKKSGHHKHQQHRQQQNKQNNNRNNQNKQQQQHHNQKQNNQRNNPNRNNNHRHPKKRKR	0.3	+0.3	-1	1	0.9	16.45
11	GQSTSTSWGGWGTSGSGGGTSSGGWSSTTSGGEDDGTDDEGTDDETADDDSTDEEGSDED	0.3	-0.3	-1	3	0.94	22.42
12	EDDDGQSTESTSWDGGWGDTSGSDGGGTDSSGGWSSTDTSGGEGTGTDTASTDGSEEDED	0.3	-0.3	-1	3	0.94	22.42
13	KKKKGHTGRTGTGRGTTGRSSGARGGSARTTTTSSSSESGSSDSSSAETASSDSSDEEEE	0.3	0	0.53	3	0.95	5.68
14	KEDKGHTGDTGTGRGTTGRSSGAKGGSAETTTTSSSSESGSSRSSSAKTASSESSERRED	0.3	0	0.13	3	0.95	7.95
15	GHTGTGTGGTTGSSGGGSTTTTSSSSSGSSSSSTSSSSDKEKDREAKERDARERERKEAAR	0.3	0	0.01	3	0.89	11.23
16	GHDTGKTGETGKGTDTGRSSEGGAGSKTTETTRSSDSSASGRSSESSRSTESSKSSEAAR	0.3	0	0.06	3	0.95	3.25
17	KRRNQKRRGTRKRSGKRKGSRKRGARRKSTATATGSTTSSATASASSSSSSSSGAGATGGS	0.3	+0.3	-1	3	0.94	13.60
18	KRRKNQGTKSGGSKGASTRATATRGSTTKSSATASASRSSSSRSSGARGATGKGSRRKRR	0.3	+0.3	-1	3	0.94	13.66
19	KKRKAHHNKNNQNRQNNNRNQNQRNQNQKNQQQDDEEEDDDDDDDEEDDDDDDDDDDEE	0.6	-0.3	0.84	1	0.95	21.22
20	DKDAHDKRHNKEDNNKDDQNEDEQNDDDNNNQDEKNQRDDNQERENQDEDNQDDDQQRDD	0.6	-0.3	0.19	1	0.97	39.66
21	KRKSNRRRPPRRKNNKKRPNRRRQQRKRNNPQDDENPEDDNNDDEQQDEEQHEDEPQEEE	0.6	0	0.64	1	0.96	-11.82
22	DDRSNERRPPERENNEKKPNEREQQDKKNNPQKRRNPDDRNNDRKQQEREQHRDEPQDRE	0.6	0	0.06	1	0.96	18.49
23	RRRKKKKTKRRKMRKRRPRKKRNKRKRPRRRKPQQNNEQQNNDNQNNENQNSDQSEDDDD	0.6	+0.3	0.72	1	0.94	4.03
24	DDRTMRRKPNKKEPPRRRQQKKDNNRRKQQNNRRENQRRDNNRKKNQKKRNSKKDQSRDE	0.6	+0.3	0.22	1	0.91	15.75
25	KKKKMVVSRSVVGRAGMARSVVAKSVVARPVSGDDDEEEEEDDDEDDDDDEEDEDDDDDD	0.6	-0.3	0.84	3	0.86	22.64
26	DDKEMVDKDVSEEKSVDEDVGKRDAGDDDMASVVDDASVDEVAPREVSGERDRDKDEDED	0.6	-0.3	0.16	3	0.92	46.66
27	KRRAVKKKVAKRKSSRKKVVKKRGARKKSVSVDEDVPEDDVVDDDGSDDDVSDDEAVDEE	0.6	0	0.64	3	0.88	-2.21
28	EKDAVDEKVADKKSSKDKVVDDRGAKRDSVSVDRDVPDEKVVRDKGSDREVSEKKAVKRD	0.6	0	0.06	3	0.9	29.24
29	SSGSSGSSEKDAVDEKVADKKKDKVVDDRAKRDVVDRDVPDEKVVRDKDREVEKKAVKRD	0.6	0	0.05	3	0.84	29.69
30	SESKGDSASVGDSESKVADKKKDKVVDDRAKRDVVDRDVPDEKVVRDKDREVEKKAVKRD	0.6	0	0.05	3	0.86	27.54
31	KRRKRKRVRKRRVKRKKAKRRRAKRRKVKKKRSAAVGEAVAADSAVADAVVVDSADDDDD	0.6	0.3	0.72	3	0.76	9.82
32	KRKVVDRDAARRRVSDRKAAKDRVGKRKAVAARKKSARKDVADEKAVRDRVVKRKSADRK	0.6	0.3	0.18	3	0.61	28.31

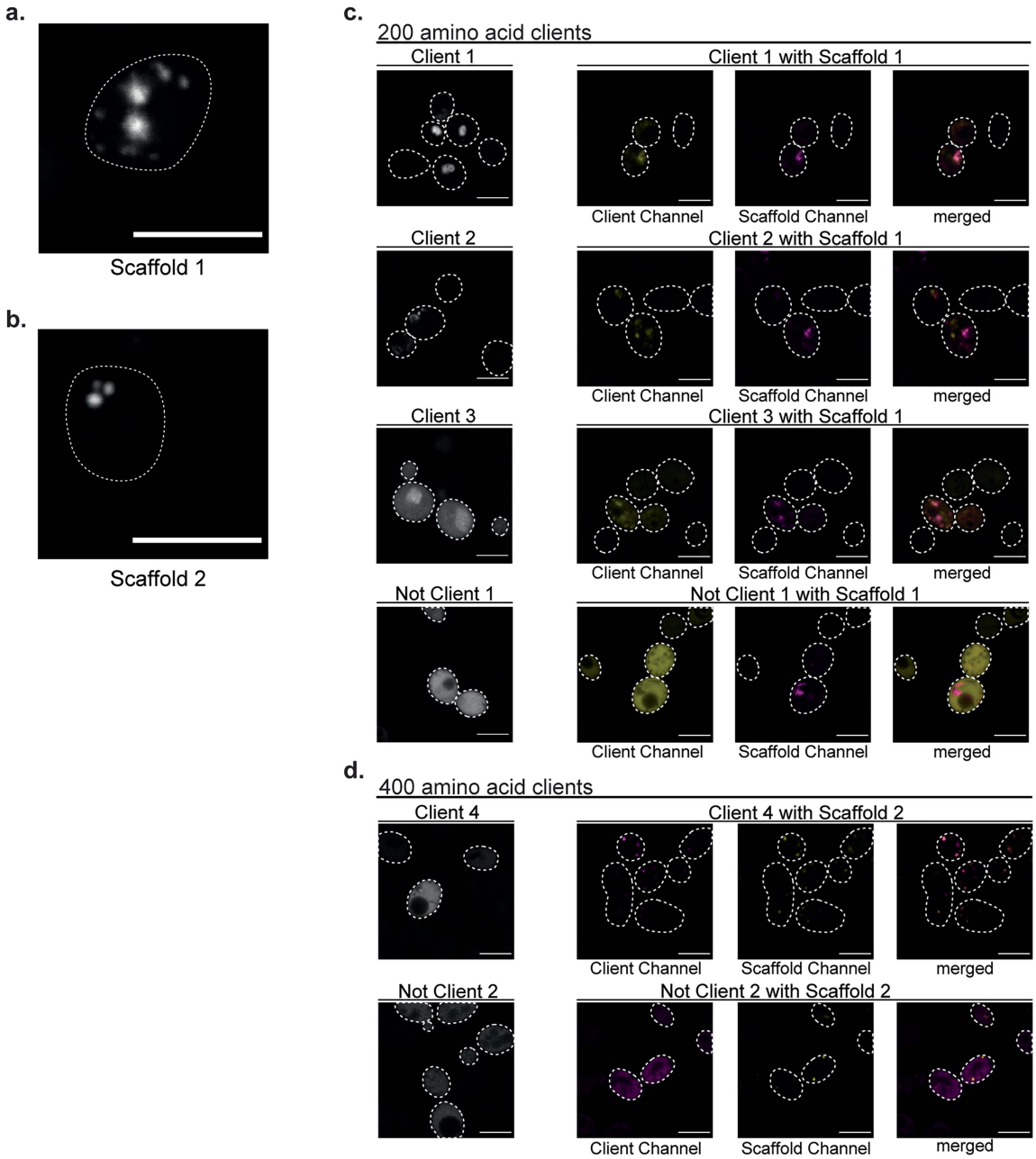
Extended Data Fig. 2 | Library variants described in Figs. 2 and 3. Columns reported here include the Fraction of Charged Residues (FCR), Net Charge Per Residue (NCPR), charge-distribution metric kappa (κ), Kyte-Doolittle hydrophobicity (H), average disorder prediction (Disorder), and a self-interaction (self ε - lower means on average stronger self-interactions). Note that a sequence must have both positive and negative residues for κ to be calculated; otherwise, it is reported as -1. Disorder here is calculated as the average disorder score as predicted by metapredict v2.0. Self-interaction score (self ε) is calculated using the CALVADOS version of the FINCHES model.



Extended Data Fig. 3 | Ensemble differs between in-cell and in silico analyses when charges are clustered. (**a**) Pairwise comparisons of in silico (blue) R_e and in cell (green) FRET efficiencies. In silico points are the mean end-to-end distance (R_{ee}) of 3 converged simulation repeats, and error bars are their standard deviation. In cell markers and error bars are the average and standard deviation of the medians of all biological repeats (n = listed in Fig. 2a). Significance is determined by an unpaired two-sided Student's t-test without multiple testing correction, where the asterisks associated with each sequence pair denote the resulting p-values: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. Sequences with evenly distributed net-positive charges all increase in ensemble dimension as the fraction of charged residues increases. (**b**) For the same sequence parameters but with clustered charges, in silico sequences display no significant ensemble change, while in cell sequences expand. This may imply environmental conditions influencing ensemble behaviour in cells. (**c**) Schematic of how GOOSE was used to generate sequences tested. All sequences here are 60 residues. Five sequences are generated with specified parameters; R_e is predicted using ALBATROSS for each sequence, and the mean is reported as the condition average. This is repeated, titrating along both the number of prolines and charged residues, until the heatmaps in panels d and e are complete. (**d**) Heatmap of average R_e as proline content and the number of charged residues increase, with evenly distributed charges. Here, as both the number of prolines and the number of charged residues increase, R_e increases. Compared sequences for which proline content differs in Fig. 2f are marked as labelled stars. (**e**) Heatmap for sequences with clustered charges. Here, there is no consistent dependence on the number of charged residues, but increases in proline content expand the ensemble. (**f**) Same format as panels a and b, but with predicted R_e instead of simulated R_e. In cell E_f markers and error bars remain the average and standard deviation of the medians of all biological repeats (n = listed in Fig. 2a). Predicted R_e markers are the mean of the condition average, while the error bars are the standard deviation (n = 5 sequences). Significance is determined by an unpaired two-sided Student's t-test without multiple testing correction, where the asterisks associated with each sequence pair denote the resulting p-values: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. For evenly distributed charges, all ensembles expand as FCR increases and/or as proline counts increase. (**g**) When charges are clustered, increasing FCR and proline count concomitantly (left) expands the ensemble under both predicted and in-cell conditions, whereas when only FCR is significantly increased (right), the ensemble compacts in cell but shows no predicted change.



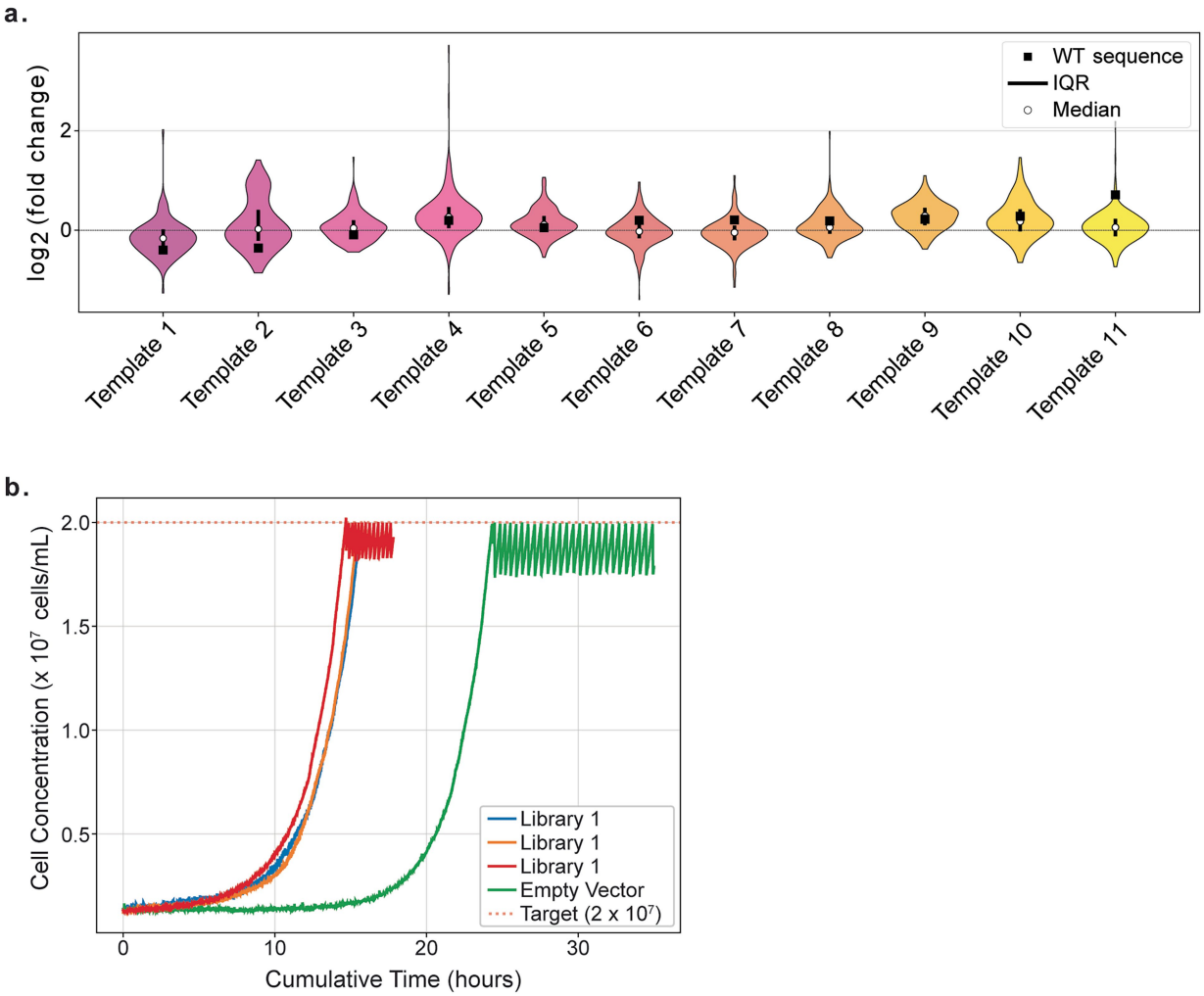
Extended Data Fig. 4 | In vitro vs. in-cell comparison of disordered protein ensembles. Pairwise E_f comparisons of in vitro (purple) and in-cell (green) data. In vitro data points are the mean of biological replicates (n = 3), and error bars are their standard deviation. In-cell markers and error bars are the average and standard deviation of the medians of all biological repeats (n = listed in Fig. 2a). Significance is determined by an unpaired two-sided Welch's t-test without multiple testing correction. Asterisks below each pair denote resulting p-values: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. (**a**) E_f for sequences with evenly distributed charges and a constant net charge of −0.3. (**b**) E_f for sequences that introduce aromatics and modulate hydrophobicity. In both comparisons, ensemble compaction is expected: overall, removing Q/N promotes compaction because of the loss of hydrogen-bonding partners. The addition of hydrophobic aromatics (W) that also engage in self-attractive pi-pi and cation-pi interactions strengthens this effect. Mutations to relatively more hydrophobic residues, S and V, would also be expected to contribute to ensemble compaction. (**c**) E_f comparisons for sequences that modulate FCR, net charge, and charge clustering. In-cell structural ensembles compact monotonically when positive charges are added, with charge patterning and FCR having little effect on this relationship. In vitro constructs behave in a more variable, sequence-dependent manner.



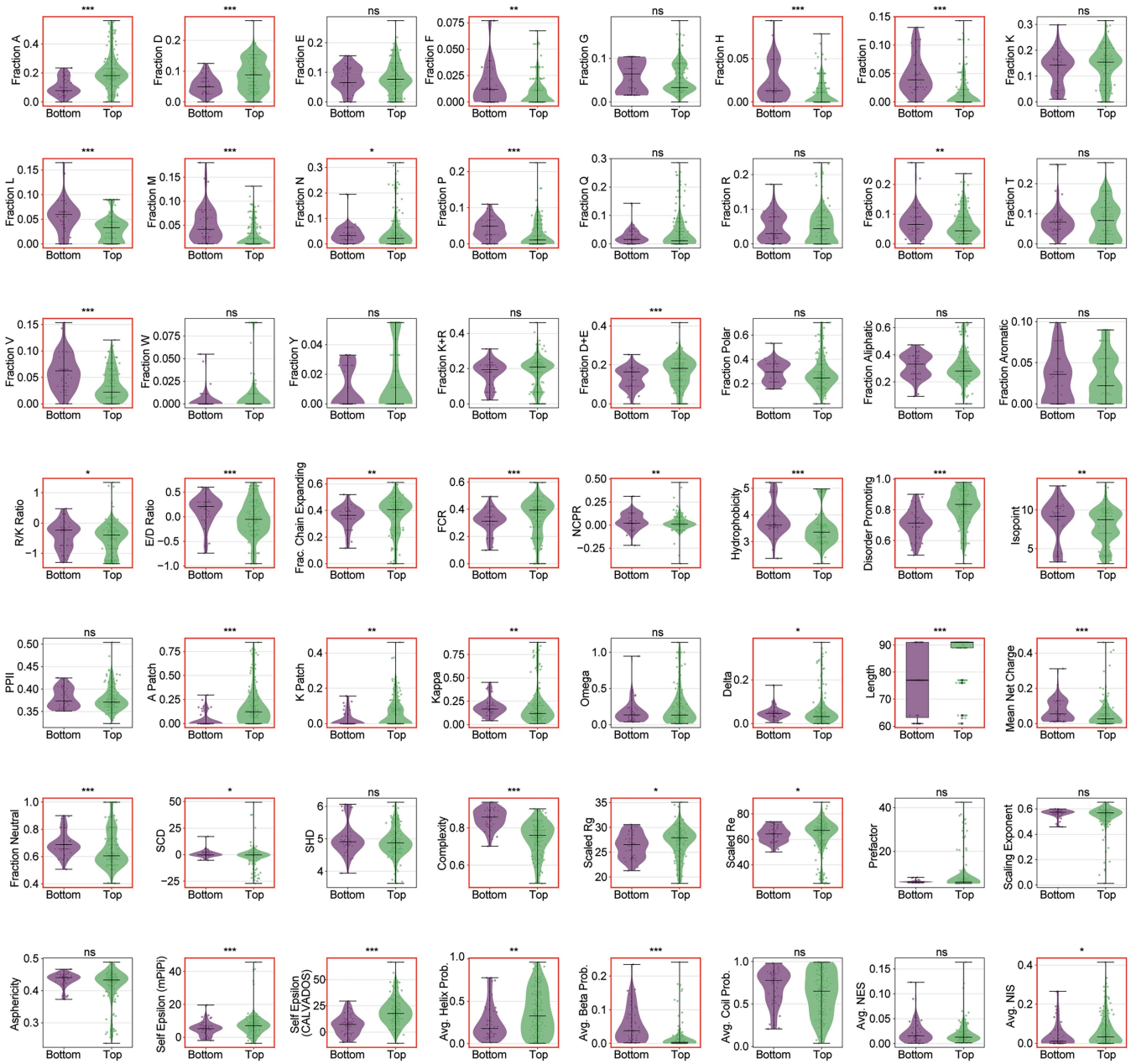
Extended Data Fig. 5 | Images of yeast expressing scaffold constructs, client constructs, or scaffold and client constructs together. a. mTagBFP2-Scaffold1 and **b**. mScarlet3-Scaffold2 alone. **c**. Image of yeast expressing client construct alone (furthest left) or client construct with scaffold (right three columns). The scaffold construct was mTagBFP2-Scaffold1. All clients were 200 amino acids and used a mScarlet3 fluorophore. **d**. Image of yeast expressing client construct alone (furthest left) or client construct with scaffold (right three columns). The scaffold construct was mScarlet3-Scaffold2. All clients were 400 amino acids in length and used the mTagBFP2 fluorophore. For all images, the scale bar is 5 μm.

#	Uniprot	Sequence	Origin	Organism	References
1	N/A	MLVAIHLLPEKSKMATIVVEKESKS PLADHAPEKSKMARVAIKVTEVHND TGEEQDKKKMM	Synthetic (deleterious)	N/A	
2	O82355	MYILKSATRTIASGTIPDPGSLVGS GTTVLDVPVKVAYSIAVSLMKDMCT DWDIDYQLDIGLTFDIPVVGDITIP VSTQGEIKLPSLRDFF	LEA (tile 7) (deleterious)	Arabidopsis thaliana (Mouse-ear cress)	https://doi.org/10.1105/tpc.114.127316 LEA_2
3	N/A	MLKDKAGSAWNQLKDKAGSAWNQLK DKAGSAWNQLKDKAGSAWNQLKDKA GSAWNQLKDKAGSAWNQLKDKAGSA WNQLKDKAGSAWNQ	LEA (motif only)	Hypsibius exemplaris (Freshwater tardigrade)	https://doi.org/10.7554/eLife.97231.3 LEA_4
4	N/A	MDATRDKLGEYKDYTADKARETNDS VARKTNETADASRDKLGEYKDYTAD KTRETKDAVAQKASDASEATKNKLG EYKDALARKTRDAKDT	Synthetic LEA	N/A	
5	P0CU49	MRPADNWAESQKEKAKAGLKDAQAE VGKVAREVKDKAAGGIEQAKDAVKQ GANDLKRSGSRTFENAKDDIQAKAQ HAKSDLKGAKHQAEGV	LEA	Hypsibius exemplaris (Freshwater tardigrade)	10.1016/j.molcel.2017.02.018 https://dx.doi.org/10.1016/j.molcel.2017.02.018
6	A0A438HAK2	MPRGYSKISDHFNRDRGQYVSKGES LRSPLKKKKNTIRGGTGKNAGEAVT ENVSNVTASARAGLKKTKATLEEKV L	Short LEA	Vitis vinifera (Grape)	LEA_1 inferred by homology
7	A0A6J3ELF5	MPTGRKQKGTACLRKLRNSASSSGR NLRNWHRTVRSSAFDVPSDLICGSS TQSEEGAPQEGILEDMPVDPDNEAY EMPSEEGYQDYEPEA	Non-Desiccation Related	Sapajus apella (Brown-capped capuchin) (Cebus apella)	Alpha-synuclein isoform X5
8	A0A1U8F7A1	METAKEKAANIAASAKSGMEKTKAI LEEKAERMSTRDPLKKEMATEKKDD KKTAAELNKREAMDQNATASGICIT S	Short LEA	Capsicum annuum (Capsicum pepper)	LEA_1
9	A3AHG5	MKEKLTVSPAATQEHLGGGEERAVK ERAAEKLGGGEERAVKERAAEKAAS VYFEEKDRLTRERAAERVDKCVEKC VEGCPDATCAHRHGKM	LEA (tile 6)	Oryza sativa subsp. japonica (Rice)	https://doi.org/10.1371/journal.pone.0045117
10	A3AHG5	MKASDASEATKNKLGEYKDALARKT RDAKDTTAQKATEFKDGVKATAQET RDATADTARKAKDATKDTTQTAADK ARETAATHDDATDKGQ	LEA (tile 4)	Oryza sativa subsp. japonica (Rice)	https://doi.org/10.1371/journal.pone.0045117
11	A0A4D6M434	MQAVKEKIVSLNATRKARQEAKAEE RAEKEIAKARMDVAREIRLAKEAEA EMDLHVANTGKKG	Short LEA	Vigna unguiculata (Cowpea)	LEA_1 inferred by homology

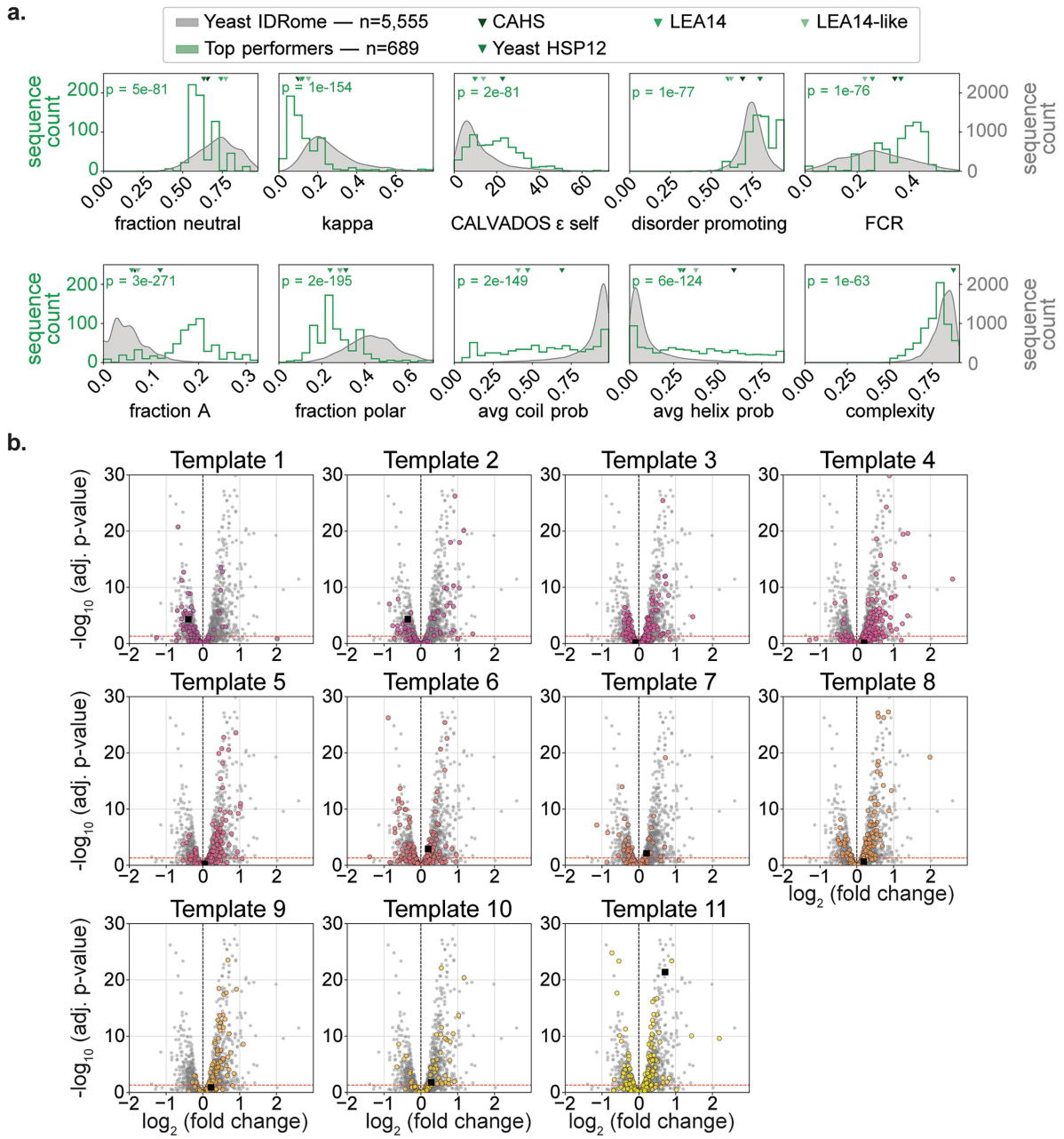
Extended Data Fig. 6 | Template sequences for protectant library reported in Fig. 5. Columns reported here include the UniProt identifier, if applicable, for each template sequence, the origin (if synthetically generated, if a LEA, and if it was a tile of a longer sequence), the organism it is found in, and references if studied or annotated.



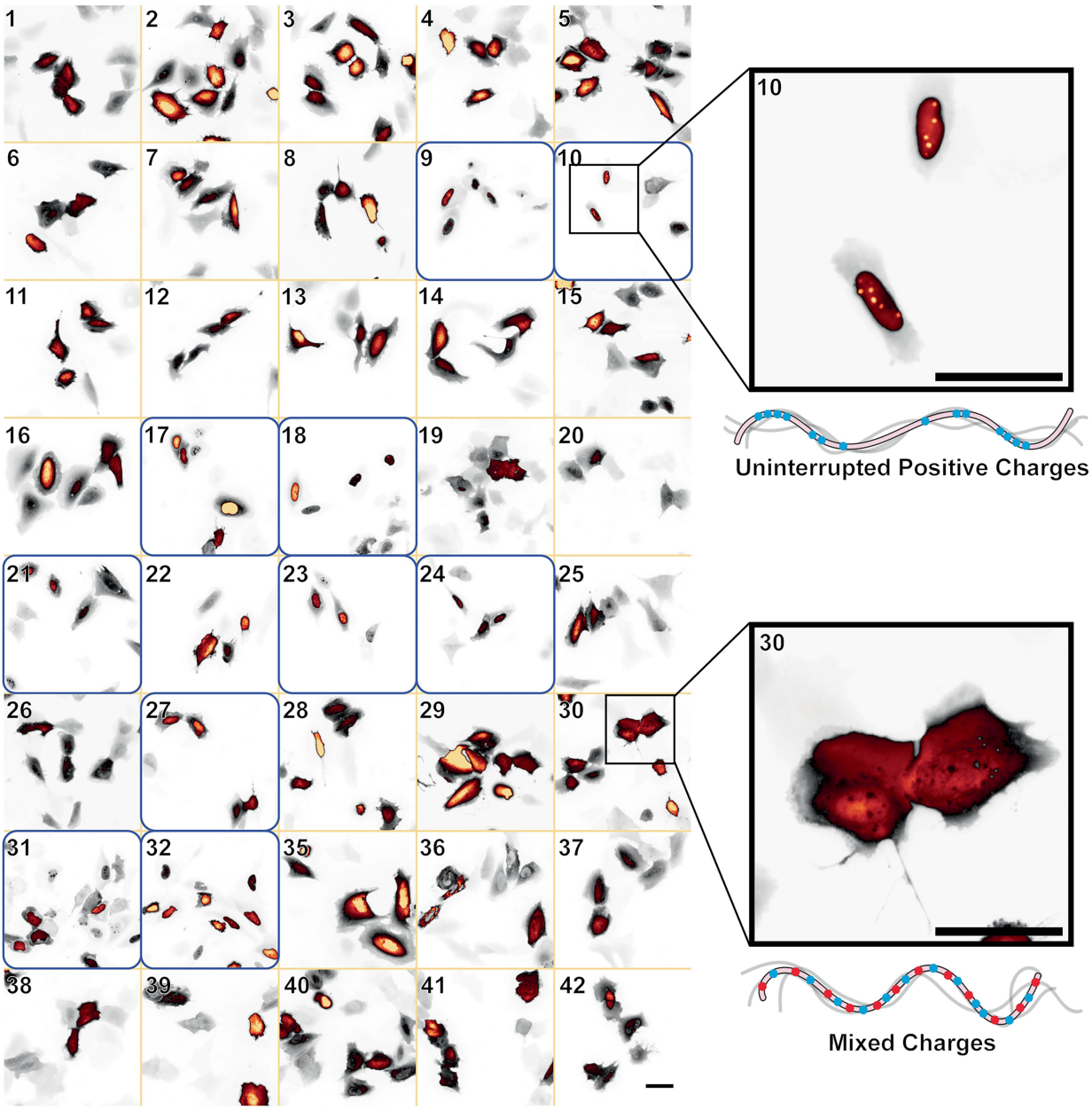
Extended Data Fig. 7 | Protection differs across templates, and the library confers faster regrowth post desiccation. a. Violin plots of all templates and their corresponding weighted log2FoldChange values across 3 separate repeats of the experiment. The wider the violin, the more frequently that specific log2FoldChange value occurs. Visualized were the WT template value, the IQR, and the median (white circle). **b**. Plotted cell concentrations of libraries (blue, yellow, red) and an empty vector control (green) over time, measured by optical density in Pioreactors. The optical density was calibrated and converted to cell concentration (cells/mL) by correlating OD with cell counts (DeNovix Celldrop FLi). The Pioreactor was programmed to remove 2 mL of cells and add 2 mL of fresh media to dilute the culture when the cell concentration reached 2.0×10^7 cells/mL. This ensures the culture remains in log phase. Removal and addition of fresh media is visualized in library 3 (red) at 15 h, and in the empty vector (EV) control at 25 h.



Extended Data Fig. 8 | Comparative distribution of top and bottom-performing sequence variants. Violin plots of the worst-performing sequences ('bottom', purple; $\log^2$(fold change) ≤ −0.5, statistically significant) and top-performing sequences ('top', green; $\log^2$(fold change) ≥ 0.5, statistically significant). N = # labelled below each category. Individual sequences are shown as black points. The middle line within each violin indicates the median; top and bottom marks indicate the maximum and minimum values. Significance between distributions of top and bottom groups was determined by a two-tailed Mann-Whitney U test, corrected for multiple testing using the Benjamini-Hochberg procedure to control the false discovery rate (FDR), and significant results ($p < 0.05$) are highlighted with a red border. Exact p-values are denoted above each plot. Sequence length is displayed as a boxplot; the horizontal line indicates the median, box edges indicate the interquartile range (IQR), whiskers extend to 1.5 × IQR, and individual points beyond this range are shown as outliers. All other sequence features are described in Supplementary Table 4.



Extended Data Fig. 9 | Comparison of sequence features to the yeast IDRome; distribution and significance of log2-fold change across sequence variants. a. Histograms of sequence features for the yeast IDRome (grey, n = 5,555) and the top-performing library sequences (green, n = 689; inverse-variance weighted $\log^2$(fold change) > 0, BH-adjusted p-value < 0.05). The yeast IDRome distribution is shown as a kernel density estimate (KDE) scaled to sequence counts using Scott's bandwidth rule, plotted on the right y-axis; top performer counts are shown as a step histogram on the left y-axis. Statistical significance between distributions was determined using a two-sided Mann-Whitney U test with Benjamini-Hochberg multiple testing correction; exact p-values are shown in the upper right of each panel for comparisons with a rank-biserial correlation |r| ≥ 0.1. Inverted triangles mark individual known desiccation-protective sequences: CAHS–106094, yeast HSP12, LEA14 (O03983), and LEA14-like (O82355). Sequence features were calculated as in Supplementary Table 4. **b.** Volcano plots of the weighted $\log^2$(fold change) and the $-\log_{10}$(BH-adjusted p-value), as calculated in Fig. 5c, highlight the original template sequence (black square) and its respective variants generated by GOOSE.



Extended Data Fig. 10 | Cellular localization and puncta formation of GOOSE-generated sequences. Representative images of the direct acceptor channel from all 40 constructs generated by GOOSE were used for E_F measurements. Numbers in the top-left of each box correspond to sequence numbers from Extended Data Fig. 2 and Supplementary Table 1. Blue boxes around sequences indicate sequences with two or more patches of three or more positive amino acids. Almost all constructs with runs of positively charged residues (excluding #27, 9/10) display nuclear puncta consistent with inclusion into RNA-ribonucleoprotein bodies, as well as strong nuclear localization (e.g., #10). When sequences lack clustered positive charges or contain mixed positive and negative charges, they display a diffuse signal (e.g., #30). Scale bar = 50 µm.

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Reporting Summary

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

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☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐ A description of all covariates tested
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☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	In-cell FRET data and miscrocopy for U-2 OS cells were taken on an Olympus IX83 microscope equipped with a laser-guided autofocus system and using a 20X 0.8 NA dry objective. Excitation light was produced with a Spectra X light engine (Lumencor, Beaverton, OR), and data were collected on two linked ORCA-fusion Gen-III sCMOS cameras (Hamamatsu Photonics, Shizuoka, Japan). All yeast microscopy data was collected using a Zeiss LSM 880 Confocal with Airyscan. In vitro FRET data was collected on a CLARIOstar plate reader (BMG LABTECH). Sequencing was done via Azenta for sample preparation and sequencing on their MiSeq platform.
Data analysis	Sequencing data analysis was done use Bowtie2 and DESeq2. Image segmentation was done using CellPose (version 4.0.4). The GOOSE software package is available at https://github.com/idptools/goose, and all documentation is available at GOOSE documentation is available at https://goose.readthedocs.io/ The KEA software package (used for DNA sequence design) is available at https://github.com/ryanemenecker/kea all other custom code used in this project is available at https://github.com/sukeniklab/GOOSE_2026, https://github.com/holehouse-lab/supportingdata/tree/master/2026/GOOSE_2026,

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Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All data to produce main text figures are provided with the paper

Data associated with main text of the figures are availalbe at https://github.com/sukeniklab/GOOSE_2026 and, https://github.com/holehouse-lab/supportingdata/tree/master/2026/GOOSE_2026,

Sequence data are available at:
https://zenodo.org/records/18774462

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	N/A
Reporting on race, ethnicity, or other socially relevant groupings	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	In-Cell FRET Microscopy Experiments: A sample size of n>205 cells per biological repeat (well) was chosen to ensure high-quality data and robust statistical comparison , while the library itself was scoped to 32 de novo sequence pairs to accommodate the relatively low-throughput nature of this precise characterization. In-Vitro FRET Experiments: Three biological replicates (derived from separate bacterial expressions) with three technical replicates per expression were used to calculate the mean and standard deviation of the in-vitro data. Coarse-Grained Molecular Dynamics Simulations: Simulations were run in triplicate from distinct starting points, utilizing a convergence threshold on the radius-of-gyration distributions to confirm sample adequacy and ensemble convergence. Yeast Desiccation-Protection Library: The size of the variant library (2,300 variants) was systematically titrated to multiplex distinct hypotheses across 11 templates. To guarantee adequate library coverage (10x coverage) and maintain statistical power, approximately 80,000 colonies were pooled, and at least 6 x 10^6 cells were utilized per experiment across three independent experimental replicates. Self-Assembly and Client Recruitment Assays: To statistically assess colocalization and partitioning behavior across unique pairs, the analysis evaluated a substantial number of individual assemblies (ranging from N = 37 to N = 91 condensates) captured across multiple cells and z-stacks.
Data exclusions	Cells excluded based on high in-cell concentrations (as assessed by direct acceptor fluorescence;) or cell death; a detailed protocol for data inclusion in microscopy is provided

Replication	For in cell FRET experiments, 6 biological repeats were performed across 2-3 different days where a biological repeat is an independent transfection event. For in vitro experiments, 3 repeats of FRET measurement were performed, where a repeat is an independent expression and purification of the construct. For yeast screening, the experiment was repeated three times, and each repeat included three biological repeats. Replications were successful when attempted.
Randomization	Randomization was not relevant to this study because the experiments were designed to systematically map exact sequence-to-function relationships rather than to compare treated versus untreated populations. Because the primary objective was to determine how specific physical and chemical parameters encode functional behavior(s) sequence variants had to be deterministically engineered to titrate a single variable while holding all other sequence properties constant.
Blinding	Blinding was irrelevant to this study due to the highly automated, script-driven nature of data collection and analysis, which eliminated subjective investigator bias. Crucial experimental metrics (in-cell and in-vitro FRET efficiencies , nuclear-to-cytoplasmic fluorescence ratios , and cellular condensate partition coefficients) were determined programmatically via defined algorithmic workflows rather than human observation.Similarly, the desiccation-protection outcomes were quantified objectively through DNA sequencing count data handled by specialized alignment software. Assessment and conclusions were never based on an individual's assessment of the data.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)	ATCC U-2 OS cells, S. cerevisiae (BY4742 for desiccation library, and BY4741 for assembly formation)
Authentication	Cells were obtained directly from ATCC and no further authentication was performed
Mycoplasma contamination	U-2 OS cells were not tested for mycoplasma contamination
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used in this work

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A